Paediatric Urology 2

Condensed rapid revision · facts already answered in Papers 1–4 point to that answer

highlighted number to memorise · **KEY POINT** and **ALGORITHM** boxes carry the reasoning · MCQ exam trivia · † added beyond the source notes · P2 topic = covered in full in that Paper's answer book

Chapter	Topic
Ch 32	Embryology of the Genitourinary Tract
Ch 33	Urologic Aspects of Paediatric Nephrology
Ch 34	Perinatal Urology
Ch 35	Urologic Evaluation of the Child
Ch 36	Paediatric Urogenital Imaging
Ch 37	Infection & Inflammation of the Paediatric GU Tract
Ch 38	Perioperative Management in Children
Ch 39	Laparoscopic & Robotic Surgery in Children
Ch 40	Clinical & Urodynamic Evaluation of LUTD
Ch 42	Bladder Anomalies in Children
Ch 43	Exstrophy–Epispadias Complex
Ch 44	Prune-Belly Syndrome
Ch 45	Posterior Urethral Valves
Ch 46	Neuromuscular Dysfunction of the LUT (Neurogenic Bladder)
Ch 47	Functional Bladder & Bowel Disorders, Enuresis
Ch 48	Lower Urinary Tract Reconstruction in Children
★	Putting it together: common paediatric pathways

32 Embryology of the Human Genitourinary Tract

1 · Overview

- Development starts in the first trimester, mostly complete by end of the second.
- **Indifferent stage** ≈ **8–9 wk**; thereafter androgens drive the sexes apart.

2 · Kidney and ureter

Three successive kidneys

Kidney	Features
†Pronephros	Cervical, week 4; non-functional, regresses
Mesonephros	†Thoracolumbar, transient; **Wolffian duct** persists. Tubules → efferent ductules, paradidymis (male); epoöphoron, paroöphoron (female)
Metanephros (permanent)	†Week 5; ureteric bud + metanephric blastema → nephrons

Reciprocal induction †

- **Ureteric bud** arises from the Wolffian duct, invades metanephric mesenchyme. Stalk → ureter; branching → pelvis, calyces, collecting tubules.
- Mesenchyme secretes **GDNF** → binds **RET** on bud → budding/branching. **WT1** makes mesenchyme competent; **PAX2** maintains bud and duct; loss → agenesis/hypoplasia (renal-coloboma syndrome).
- Bud tips induce mesenchymal-to-epithelial transition → renal vesicle → comma → S-shaped body → nephron. Nephrogenesis ends ≈ 34–36 wk; none after birth.

> **KEY POINT**
> One defect gives both dysplasia and an abnormal orifice: bud position on the Wolffian duct sets where it meets the blastema (kidney quality) and where its orifice lands in the bladder.

Ascent and rotation †

- Pelvis → lumbar (weeks 6–9), taking successively higher arteries; hilum rotates 90° anterior → medial.
- Failure → pelvic/ectopic kidney, malrotation, multiple arteries; lower poles fuse → **horseshoe kidney**, trapped under the inferior mesenteric artery.

3 · Cloaca, bladder, trigone and urethra

- P2 Embryogenesis of urinary bladder — *urorectal septum, vesicourethral canal vs UGS, wall layers, trigone (apoptosis, mostly bladder muscle), urachus, Weigert–Meyer*
- Anorectal malformations arise from the **cloacal membrane**, not the urorectal septum (weeks 5–6).
- Bladder/urethra alike in both sexes except the larger male prostatic urethra.

Weigert–Meyer rule (duplex kidneys)

- P3 Embryology and management of ectopic ureter. — *bud too low → lateral refluxing lower pole; too high → caudal/medial ectopic upper pole; male vs female ectopic sites and sphincter*
- Upper-pole orifice rotates dorsally relative to lower-pole. Male ectopia stays above external sphincter → no continuous wetting; †female below → continuous dribbling with normal voids.

- **Ectopic vas**: vas joins the ureter instead of the verumontanum; both drain into the prostatic urethra.

4 · Gonads

- P1 Testicular development, gonadal descent, abnormalities of descent and bilateral undescended testis — *indifferent gonad, PGC migration, SRY → Sertoli cords, rete/efferent ductules, AMH/testosterone/INSL3 table, remnants*
- Primary sex cords week 6; human seminiferous cords appear to derive from mesonephric tubules.
- **SRY** → Sertoli cells enclose PGCs in cords (≈ **6 wk**). Leydig cells from 8 wk.
- **MIS/AMH** (Sertoli): Müllerian regression **8–10 wk** (sensitivity window 9–13 wk); absent → uterus, tubes persist. **Testosterone** (Leydig): Wolffian → epididymis, vas, seminal vesicle; absent → Wolffian ducts regress.

Ovary — the default

- No SRY → no Sertoli, MIS or androgen → female. Cortex/medulla ≈ 8 wk; germ-cell **nests** 10–11 wk.
- FOXL2+ cells → primordial follicle → primary (≈ 20 wk) → secondary (term) → antral (postnatal). Oocytes arrest in meiosis I until puberty.

> **KEY POINT**
> Two independent testicular signals act on two independent duct systems: MIS decides Müllerian, testosterone decides Wolffian and external genitalia. Every DSD is a combination of the two.

5 · Gonadal descent

- P1 Testicular development, gonadal descent, abnormalities of descent and bilateral undescended testis — *CSL + gubernaculum; transabdominal (10–15 wk: androgen, INSL3) and inguinoscrotal (7th month: genitofemoral nerve CGRP) phases; female ligaments*
- *Insl3*-null mice: isolated intra-abdominal cryptorchidism; gubernacular swelling androgen-independent, inguinoscrotal phase androgen-dependent.

6 · Genital ducts

Male accessory glands

- **Prostate**: buds at **10 wk** from FOXA1+ UGS epithelium below ureteric insertion; most buds lateral to the **verumontanum** (Wolffian ducts + utricle open on it).
- Androgen acts via **AR in mesenchyme**; stromal 5αR2 makes **DHT (~10× AR affinity)**. 5αR2 deficiency → rudimentary prostate + severe hypospadias. Mesenchyme can induce prostate even from bladder epithelium.
- Oestrogen → squamous metaplasia via ESR1 in stroma **and** epithelium.
- **Seminal vesicle**: Wolffian diverticulum 10–12 wk; main anomaly = cyst.

Female tract

- Week 5: Müllerian ducts use the **Wolffian duct as guidewire** to reach the UGS (hence combined anomalies).
- Week 8 fuse → uterovaginal canal; septum gone by week 9. Unfused cranial ducts → tubes; canal → corpus, cervix.

- **Vagina**: PAX2+ Müllerian plate replaced (14–21 wk) by FOXA1+ UGS epithelium → **entire vaginal epithelium and hymen UGS-derived** (Bulmer over Koff). Canalisation 15–19 wk.
- Female Wolffian remnants: epoöphoron, paroöphoron, **Gartner duct**.

7 · External genitalia

- P3 Embryology of the penis and management of proximal hypospadias (including Koyanagi repair) — *anogenital distance first, germ layers, double zipper (opening androgen-independent, closing dependent), glanular canalisation, timetable, 12 o'clock nerve-free zone*
- Shaft fused by **16 wk**. Meatus = ecto–endodermal junction; prepuce depends on glanular urethra.
- Tunica thickest at the 12 o'clock nerve-free zone → dorsal plication for mild–moderate curvature.
- **Female**: vestibular plate canalises, folds do not fuse → open vestibule; clitoris = penis without urethra, same 12 o'clock zone.

8 · Developmental step → anomaly

Step	Anomaly
†No bud / no induction (GDNF–RET, PAX2)	Renal agenesis (absent hemitrigone)
†Abnormal bud–blastema contact	Dysplasia, hypoplasia, MCDK
†Early bifurcation / two buds	Bifid ureter / complete duplication
Bud too low / high	VUR / ectopic ureter
†Ascent/rotation fails	Pelvic, malrotated, horseshoe kidney
Cloacal membrane / septation	Anorectal malformation; †cloacal anomalies, exstrophy
Allantois persists	Urachal remnants
Fold fusion arrests (shaft–glans junction commonest)	Hypospadias
Urethral plate splits (Sonic hedgehog)	Urethral duplication (complete from bladder; Y along shaft)
Androgen excess in 46,XX	Vestibular fusion → virilisation (CAH)
CSL retained	Intra-abdominal testis
Gubernacular attachment abnormal	Canalicular / ectopic testis (perineal, femoral, prepenile)
Gubernacular migration fails	Congenital cryptorchidism
Processus fails to close / involute	Hydrocele, hernia / acquired (ascending) testis
One Müllerian duct absent	Unicornuate uterus
Ducts fail to fuse	Uterus didelphys
Septum persists	Uterine obstruction/duplication
Vaginal plate fails to canalise	Transverse vaginal septum (middle–upper third)
Inferior end fails to open	Imperforate hymen — commonest vaginal obstruction
Müllerian + Wolffian abnormal	OHVIRA

9 · Clinical correlations

- **Hypospadias** (≈ 1 in 250): ectopic meatus, dorsal hood, ventral skin deficiency ± curvature. Abortive spongiosal elements alongside → **arrest, not deformation**. Cause mostly unknown (genetics + endocrine disruptors).
- **CAH**: autosomal recessive → 46,XX virilisation, may look fully male. Clues: **impalpable gonads + hyperpigmentation**; mimics hypospadias with bilateral impalpable testes.
- **Streak gonad** (45,X; 46,XX or 46,XY dysgenesis): no MIS/testosterone → female. Hormone replacement; **remove streak if 46,XY** (gonadoblastoma).
- **Ovotesticular DSD**: ducts follow ipsilateral gonad; female fertility possible, spermatogenesis not reported.
- **PMDS**: 46,XY with uterus/tubes (hernia uteri inguinale), usually MIS-receptor defect; found at exploration for bilateral abdominal testes.
- **OHVIRA**: post-menarche cyclic pain, discharge, pyocolpos. Kidney often **dysplastic with ectopic ureter into the obstructed hemivagina**, not agenesis → persistent leak after septum resection if missed.

> **KEY POINT**
> Homologues: Littre ↔ Skene, Cowper ↔ Bartholin, spongiosum ↔ vestibular bulb, gubernaculum ↔ ovarian + round ligaments, labioscrotal swellings ↔ labia majora. Traps: vagina is UGS-derived; appendix testis Müllerian, appendix epididymis Wolffian.

33 Urologic Aspects of Pediatric Nephrology

1 · Renal function in the child

- CAKUT = **commonest cause of paediatric ESKD**.
- GFR doubles in 2 weeks, adult level by **2 years**; term ≈40 mL/min/1.73 m².
- Creatinine: first 48 h = **maternal**; by day 7 the neonate's own (**<0.5 mg/dL** at term).
- †Schwartz: eGFR = 0.413 × height (cm) / creatinine (mg/dL).
- Neonate dilutes to 50 mOsm/L, concentrates poorly; bicarbonate matures over 3–4 weeks.

> **KEY POINT**
> The neonatal kidney is "hormonally deaf": maternal creatinine, week-1 salt wasting, hyperkalaemia (aldosterone-unresponsive tubule), low bicarbonate and poor concentration (ADH-unresponsive) reflect immaturity, not disease.

2 · Haematuria

- Microscopic: **>5 red blood cells (RBCs)/high-power field** (fresh spun); ≈1% of school children, mostly resolves.
- **Fresh microscopy is critical**: red → non-glomerular; brown "tea-coloured" + RBC casts → glomerular. Dipstick-positive, no RBCs → haemoglobin/myoglobinuria.
- **UTI = commonest cause of gross haematuria** → culture. Trauma, pain, clots → urological work-up (often CT). Stones → *Ch 54*.
- Suspected GN: creatinine, albumin, C3/C4, ASO/anti-DNase B → nephrology.
- Isolated microscopic haematuria: US low-yield; **VCUG and cystoscopy not indicated**.
- Gross-only: urethrorrhagia, Wilms/bladder tumour, renal vein thrombosis, papillary necrosis, nutcracker. Microscopic-only: thin basement membrane, interstitial nephritis, sickle cell.

3 · Proteinuria

- Normal UPCR **<0.2 mg/mg** (<0.7 newborn); timed <4 mg/m²/h; adolescents <150 mg/24 h.
- **Transient** (fever, exercise) >75% of isolated cases. **Orthostatic**: normal first-morning UPCR — benign. **Fixed** (glomerular, tubular, reflux, obstruction, dysplasia) → full evaluation ± biopsy.

4 · Glomerular disease

Age: <1 year → genetic (Denys-Drash, Frasier); 1–10 → minimal change, post-streptococcal GN, HSP, HUS; ≥7 → lupus, IgA, MPGN, Alport.

Nephrotic syndrome

- Proteinuria **>40 mg/m²/h or UPCR >2.0** + hypoalbuminaemia + hyperlipidaemia.
- Congenital (<3 months): immunosuppression unhelpful. Primary: **steroids first, no biopsy** — minimal change is 90% under 10 years, ~90% steroid-sensitive.

Category	Definition	Action
Steroid-sensitive	Remission within 8 weeks	~50% relapse

Category	Definition	Action
Frequently relapsing	≥2 relapses in 6 months or ≥4 in 12	Steroid-sparing: cyclophosphamide, mycophenolate, rituximab, calcineurin inhibitors
Steroid-dependent	2 consecutive relapses on steroids	
Steroid-resistant	No remission after 6–8 weeks	Biopsy → usually FSGS (recurs after transplant)

Nephritic syndrome

Haematuria + proteinuria + hypertension + AKI. Complement:

C3	C4	Diagnoses
Low	Normal	Post-infectious GN, C3 glomerulopathy
Low	Low	Lupus nephritis, MPGN
Normal	Normal	IgA, HSP, Alport, ANCA GN, anti-GBM

- **Post-streptococcal GN**: weeks after pharyngitis/pyoderma; C3 normalises by **8–12 weeks**; persistently low → biopsy.
- **IgA nephropathy**: commonest primary GN; painless gross haematuria **concurrent with** respiratory infection.
- **Alport**: type IV collagen (mostly X-linked COL4A5); haematuria + high-frequency sensorineural deafness; ~50% of males ESKD by 25.
- **HUS**: commonest cause of severe AKI in a healthy young child. Shiga-toxin E. coli → supportive, ~half dialysed. Atypical (complement) → eculizumab.

5 · Tubulopathies

- P2 Renal tubular acidosis — *types 1/2/4 compared: defect, urine pH, K^+, stones, causes (obstruction → type 4), treatment*
- Fanconi adds glucosuria, aminoaciduria, phosphaturia (cystinosis, drugs). Distal RTA: hypercalciuria, **nephrocalcinosis/stones**, rickets. Type 4 (**hyperkalaemia**) from obstructive uropathy → urological correction.

6 · Cystic disease

- **MCDK**: non-communicating cysts, non-functional; bilateral lethal. VUR or PUJO in up to 25% → VCUG, especially with hydroureteronephrosis. US/BP 3-monthly in year 1 → 6-monthly to involution or ≥5 years → annual into adulthood.
- **ARPKD** (*PKHD1*): enlarged kidneys, hepatic fibrosis; >75% ESKD by 5 years. ADPKD (*PKD1/PKD2*): usually adult.
- Nephronophthisis: polyuria, anaemia → ESKD by adolescence (Senior-Løken, Joubert, Bardet-Biedl).

7 · Hypertension

- Repeated BP **>95th percentile** for age, sex, height; adolescents normal <120/80. Up to a quarter white-coat.
- Cause by age: infants → coarctation/renovascular; early childhood → Wilms, neuroblastoma; school age → **renal parenchymal incl. reflux nephropathy**; obese >6 years/adolescents →

essential.

- No screening US >6 years unless urinalysis/function abnormal. Echo when drugs start.
- Lifestyle for stage 1 without end-organ damage; drugs for stage 2, symptoms, end-organ damage, secondary cause. Target <90th percentile; CKD **<50th**.

8 · Acute kidney injury

KDIGO stage	Creatinine	Urine output
I	↑ ≥0.3 mg/dL in 48 h or 1.5–1.9×	<0.5 mL/kg/h for 6–12 h
II	2.0–2.9×	<0.5 for ≥12 h
III	≥3× or ≥4.0 mg/dL or clearance <35 (<18 years)	<0.5 for ≥24 h or <0.3 for 12 h

- Creatinine lags GFR by 24–72 h; rises only after ~50% loss.
- FeNa **<1%** = prerenal; >3% = intrinsic; on diuretics FeUrea <35%.
- Prerenal → isotonic fluid, no K^+ until urine flows. Dose drugs as GFR <15.
- RRT for diuretic-resistant overload, refractory hyperkalaemia/acidosis, uraemia, nutrition. Two-thirds later CKD.

9 · Chronic kidney disease

- Abnormality >3 months; stages (>2 years) by GFR: I ≥90, II 60–89, III 30–59, IV 15–29, V **<15** or dialysis.
- Young: CAKUT, cystic; adolescents: glomerular.
- Complications: growth failure (GH **resistance**), bone disease, anaemia (EPO + iron), acidosis, cardiovascular death.
- **Do not restrict protein**; strict BP control (50th percentile) is the only proven way to slow progression.

10 · Dialysis and transplantation

- P1 Urological evaluation and management of pediatric kidney transplant patient. — *lower-tract evaluation and reconstruction before transplant, native nephrectomy indications, ureteric techniques, complications*
- PD first if **<9 years / <20 kg**; vesicostomy acceptable; contraindicated by omphalocele, gastroschisis, diaphragmatic hernia, exstrophy.
- Often pre-emptive living-related; many wait for **≥10 kg and 65 cm** (graft thrombosis).
- Living donor: delayed graft function 5% vs 15%; half-life 20 vs 12.5 years. Risks: CMV, EBV, BK virus; malignancy ≈7-fold.

34 Perinatal Urology

1 · Fetal renal development and prenatal imaging

- P3 Antenatal hydronephrosis (UTD) and fetal uropathy (LUTO) — *UTD 1–3% of pregnancies, second after cardiac; 18–20 wk scan; APD, SFU and UTD grading; A1/A2–3 and P1–P3 tables; causes with incidence; postnatal plan*
- Metanephric kidney at **day 28**; urine production from **14 weeks**; after **~16 weeks** amniotic fluid is mainly fetal urine → oligohydramnios → pulmonary hypoplasia.
- AFI (four largest pockets): normal 8–18 cm; oligohydramnios **<5-6 cm**.
- US severity tracks PUJ obstruction and PUV, not VUR or duplex. Exstrophy and neural tube defects usually diagnosed correctly; genital ambiguity not.
- Fetal MRI only for anatomy: cloaca, urogenital sinus, exstrophy, duplex with outlet obstruction.

KEY POINT
Prenatal detection is for triage and counselling (differential, postnatal plan, natural history); no proof it improves outcome.

2 · Grading systems

- **APD**: transverse mid-kidney at hilum (error: sagittal). Rises with GA → **meaningless without its GA**. Severe = >10 mm (2nd trimester), >15 mm (3rd).
- Postnatal pathology ≈12% (mild) → ≈88% (severe); **VUR shows no trend**.
- **SFU** (postnatal): 1 sinus barely split; 2 major calyces; 3 minor calyces, parenchyma spared; 4 parenchymal thinning. All SFU 3 keep DRF >40%; ~70% of SFU 4 do. Weakness: subjective, ignores ureter/bladder.
- Pyramidal thickness **≤3 mm** predicts pyeloplasty and reduced function.
- **UTD** (six parameters + oligohydramnios). Normal APD ceiling **<4 mm (16–27 weeks) → <7 mm (≥28 weeks) → <10 mm (postnatal >48 h)**, all else normal. Objective + subjective; pre- and postnatal; whole tract.

3 · Prenatal follow-up

ALGORITHM
A1 (APD 4 to <7 mm at 16–27 wk; 7 to <10 mm ≥28 wk; all else normal) → repeat US after 32 weeks → postnatal US after 48 h
A2–3 (any abnormal parameter) → US every 4 weeks to delivery (may progress; second anomaly likelier) → postnatal US after 48 h

- Multidisciplinary prenatal consult: exstrophy, cloaca/urogenital sinus, ARPKD, outlet obstruction (PUV, PBS, urethral atresia), myelomeningocele.

4 · Fetal LUTO and oligohydramnios

- Male fetus, bilateral hydroureteronephrosis, non-cycling bladder, **"keyhole" sign**, oligohydramnios. Only fetal cystoscopy separates PUV, urethral atresia, PBS.
- "Pop-offs" (urinary ascites, urinoma, VUR, unilateral dysplasia) mark a system under tension.

- Poor prognosis: oligohydramnios **before 27 weeks**, cortical cysts, echogenicity. After 27 weeks → favourable lungs.
- Fetal urine biomarkers **not validated** (European consensus 2022). †Classic favourable serial values: Na <100 mEq/L, Cl <90 mEq/L, osmolality <200 mOsm/L, β2-microglobulin <4 mg/L.

Ruano stage	Fluid / kidneys	Urine biochemistry	Intervention
I — mild	Normal fluid and echogenicity	Favourable	Not indicated
II — severe, function preserved	Oligo/anhydramnios; hyperechoic; no cysts	Favourable within 3 samples	**Indicated** — lungs and kidneys
III — severe, function abnormal	Anhydramnios; hyperechoic ± cysts, dysplasia	Unfavourable after 3 samples	Maybe — lungs only

Options

- Early delivery, vesicoamniotic shunt (VAS), fetal cystoscopy, serial amnioinfusion. Window **18–24 weeks**; need ≈1 in 60,000.
- Complications ≈**44%**: shunt failure (commonest), preterm labour, membrane rupture, ascites, chorioamnionitis, bowel herniation.

Feature	VAS	Fetal cystoscopy
Diagnoses cause	No	Yes — views posterior urethra
Mechanism	Bladder → amniotic bypass	Laser valve ablation → bladder cycles
Specific complications	Blockage, migration, bowel herniation	Urethrocutaneous/rectourethral fistula (≤10%)
Survival / renal function (PUV)	Improved / not improved	Improved / short-term only

- VAS before 16 weeks gave better lung/renal outcome in one series; newer shunts safe even before 14 weeks.
- Serial amnioinfusion: lungs in bilateral renal agenesis (no urine to shunt); RAFT trial ongoing.
- After VAS: over half renal insufficiency, a third RRT, a third CIC.

> **KEY POINT**
> Counsel: intervention improves perinatal survival (lungs); complications exceed 40%; it does not reliably prevent renal replacement — dysplasia is often established first.

5 · Postnatal imaging

- **First US after 48 h** (neonatal dehydration under-reads dilation); delay up to 2 weeks harmless.
- Postnatal P1 (APD 10 to <15, central calyces) / P2 (≥15, peripheral calyces or abnormal ureter) / P3 (abnormal parenchyma or bladder) ≈ SFU 1–2 / 3 / 4. **Classify by highest-risk feature**: parenchymal thinning = P3 at any APD.
- **VCUG**: VUR up to 31%; up to 25% with VUR have normal first US. Order by **UTI risk** — females, uncircumcised males, high-grade dilation, dilated ureter, abnormal bladder. ALARA, non-balloon catheter, cycling. Outpatient unless outlet obstruction suspected (bladder wall ≥0.5 cm, distended bladder, bilateral hydroureteronephrosis). Surveillance of known VUR safe; benefit of finding asymptomatic VUR unproven.
- CeVUS: no radiation, possibly more sensitive for VUR, poor urethral view.

- **MAG3** (over DTPA; "well-tempered" protocol): surgery considered if **DRF <40%**, T½ >20 min, or delayed upright retention. Pitfalls: dehydration, furosemide before peak uptake into a capacious pelvis, ROI placement.
- **DMSA**: dysplasia/scars in VUR, MCDK non-function, duplex upper-pole function. **MRU**: ectopic ureter, megaureter, crossing vessels; needs sedation.

6 · Postnatal management, UTI and prophylaxis

- P1: US 3–6 months, no CAP/VCUG/MAG3; resolves by 2–4 years. P2: CAP at discretion, **yes if ureter >7 mm**; VCUG/MAG3 at discretion. P3: US ~1 month, CAP, VCUG, MAG3 if persistent.
- Dissent: reflux raised at every grade → some base VCUG on ureteral dilation or LUTO features.
- CAP without VUR in P3 until pyeloplasty or **1 year** (~90% of UTIs in year 1).
- UTI 8–22%. Agreed risks: **female sex, high-grade dilation, intact foreskin**; debated: ureteral dilation, VUR, no CAP.
- Circumcision: NNT **10**; OR ≈0.3.
- CAP evidence weak; target high-grade UTD, females, uncircumcised, dilated ureter/VUR. SFU registry (low-grade, UTI 8%) no benefit vs McMaster (high-grade, 19%) benefit — case mix.
- Agents: amoxicillin, first-generation cephalosporin from birth; nitrofurantoin/trimethoprim **≥8 weeks** (immature hepatic metabolism).

> **ALGORITHM**
> US after 48 h → normal (<10 mm, all normal) → transient
> P1 → US 3–6 months → P2 → CAP if ureter >7 mm; VCUG/MAG3 at discretion → P3 → CAP + VCUG + US ~1 month → persistent → MAG3
> Bilateral severe dilation ± oligohydramnios → outlet obstruction: decompress → CAP → urgent evaluation

7 · Specific antenatal diagnoses

- **Transient** (50–70%, commonest): mild resolves in 90–98%; SFU 1 by ~2.5 years, SFU 2 ~95% by 4 years. Warn: VUR possible.

PUJ obstruction pattern

- Commonest **surgical** cause. Third-trimester APD **15 mm**: sensitivity/specificity ≈80% for pyeloplasty; postnatal APD predicts better. Even severe UTD: ~a fifth need surgery.
- VCUG excludes VUR and checks posterior urethra in boys; MAG3 at 1 month for persistent P3, or serial US (P2, APD <20 mm, PT >3 mm → DRF >40% likely).
- Surgery: rising dilation, DRF <40% with poor drainage, falling DRF, upright retention, symptoms → *Ch 53*.

Vesicoureteral reflux

- Prenatally detected: male, high-grade, bilateral, congenital dysplasia (up to 40%), often early resolution. Febrile-UTI presentation: female, lower grade, acquired scarring. Screen girls with high-grade UTD and uncircumcised boys; not circumcised boys with low-grade UTD.

Megaureter / UVJ obstruction

- Obstructed, non-obstructed, refluxing, refluxing-obstructed. Exclude single-system ectopic ureter.
- Most resolve. Surgery predicted by ureter **>13 mm**, DRF <30%.
- BAPU: prenatal ureter **>7 mm after 30 weeks** → postnatal US, CAP, VCUG for all; consider MAG3.
- Surgery for DRF <40%, febrile UTI, pain. **<1 year → temporise** (cutaneous ureterostomy, pyelostomy, refluxing reimplant); reimplant after 1 year †(tapered ureter in small bladder obstructs). Follow into adulthood.

> **KEY POINT**
> In PUJ, VUR and megaureter, dilation alone is not an indication: operate for function loss, infection or progression; grade sets surveillance.

LUTO conditions

- **PUV**: commonest LUTO (1.4–2.2/10,000) *→ Ch 45*. **PBS**: megacystis + deficient abdominal wall + impalpable testes; ± urethral atresia/megalourethra *→ Ch 44*.
- **Urethral atresia**: catheter will not pass → vesicostomy → PADUA (progressive augmented dilation of anterior urethra).
- **Megacystis (massive reflux)**: lateral orifices, urine cycles bladder ↔ ureters; no oligohydramnios.
- **MMIHS**: distended non-obstructed bladder; vesicostomy or CIC; CAP only for recurrent UTI.

> **KEY POINT**
> Bilateral dilation with normal fluid = urine still leaves the fetus (reflux, megacystis); with low fluid = true outlet obstruction.

Neonatal management of LUTO

> **ALGORITHM**
> Team at delivery → (lung resuscitation) → 5 Fr feeding tube, finger in rectum to guide over high bladder neck → KUB confirms → fails → US-guided suprapubic catheter

- **No balloon catheter**: balloon obstructs posterior urethra and compresses ureteric orifices.
- No immediate VCUG with renal insufficiency; decompression guided by **creatinine nadir**. Warn of post-obstructive diuresis. Oliguria after decompression → thick bladder obstructing ureters → antimuscarinics.

Multicystic dysplastic kidney

- Non-communicating cysts, no central cyst. VUR ≈20%; contralateral PUJ obstruction.
- **No Wilms association** → no routine nephrectomy. No CAP; VCUG/DMSA optional. Serial US, BP, proteinuria through puberty. Solid component → remove.

Duplex, ureterocele, ectopic ureter

- **Meyer–Weigert**: upper pole inferomedial → obstruction/ectopia; lower pole superolateral → reflux ("drooping lily").
- Suspect duplex when ureter more dilated than pelvis. Ureterocele = thin-walled intravesical cyst; in girls can cause LUTO.
- Examine introitus, US with bladder, CAP from birth, VCUG, scintigraphy.

- Endoscopic puncture (increasingly first-line; mandatory for outlet obstruction/sepsis); upper-tract approach if no VUR; high-grade VUR → usually ureterocele excision/common-sheath reimplant.

Exstrophy

- Both: non-visualised bladder, low umbilicus, pubic diastasis. **Cloacal (OEIS)** adds omphalocele, imperforate anus, spinal/renal/limb anomalies → KUB, renal + spinal US ± MRI. Bladder exstrophy: renal US, not an emergency; detected prenatally ~50%.

Cloaca, urogenital sinus, CAH

- Bladder + vaginal distension = outlet obstruction; reflux via tubes → urinary ascites.
- **Cloaca** (~1/50,000 girls): single channel, pelvic cyst, calcified meconium; polyhydramnios → suspect TOF. **Urogenital sinus**: no bowel obstruction, normal external genitalia.
- **CAH**: 21-hydroxylase (6p), AR; salt-wasting 75%. Only **1 in 8** at-risk pregnancies is an affected girl → prenatal treatment controversial. Electrolytes first; 17-OHP + genetics.

Myelomeningocele

Fetal closure before 26 weeks: less VP shunting, better motor/cognition; risks preterm labour, uterine dehiscence; **no bladder benefit**.

8 · Neonatal urological emergencies

Urinary ascites and retention

- Urine = up to 30% of neonatal ascites (respiratory distress, hyponatraemia, hyperkalaemia, ↑creatinine). **PUV commonest**; also bladder rupture (spontaneous heals with catheter; umbilical catheter injury → repair).
- Retention >24 h: boys → PUV, atresia; girls → urogenital sinus, ureterocele. Catheter fails → angiocatheter (16 G → 3.5 Fr; 14 G → 5 Fr). Persistent distension → vesicostomy ± vaginostomy.

Abdominal masses

- Solid renal: **congenital mesoblastic nephroma** commonest <6 months (hypertension, polyhydramnios); Wilms 1–5 years. Ovarian cysts mostly resolve by 6 months.
- **Adrenal haemorrhage vs neuroblastoma**: no Doppler flow vs vascular; shrinks vs persists on 1–2-weekly US; eggshell vs stippled calcification. Haemorrhage right-sided, bilateral 10–15%; may track via processus to mimic acute scrotum.

Anorectal malformation and ambiguous genitalia

- ARM: GU/spinal anomalies up to **75%** → renal + spinal US; CAP until videourodynamics at 6 weeks.
- Ambiguous genitalia: DSD team; defer gender assignment. Commonest CAH (46,XX) — impalpable gonads, uterus. **Exclude salt-wasting first**; karyotype, electrolytes, 17-OHP, testosterone, DHT → *Ch 59*.

Interlabial masses

Mass	Position	Management
Imperforate hymen	Midline bulging membrane	Hymenal incision
Urethral prolapse	Midline collar at meatus	Conservative
Prolapsed ureterocele	Off-midline from meatus; hymen intact	US + VCUG

Mass	Position	Management
Paraurethral cyst	Off-midline, displaces meatus	Most resolve

Gross haematuria

Exclude oestrogen withdrawal bleed, urate crystals → Doppler US. Haematuria + hypertension = severe disease.

Feature	Renal vein thrombosis	Renal artery thrombosis	Cortical necrosis
Risk	Maternal diabetes, prematurity, UVC, polycythaemia; prothrombotic ≤50%	Umbilical artery catheter	Hypoperfusion
Presentation	**Triad**: haematuria + flank mass + thrombocytopenia	Hypertension, heart failure	Renal failure
Treatment	Hydration ± anticoagulation	Remove catheter ± thrombolysis	Supportive

Scrotal masses and torsion

- Differential: extravaginal torsion, hernia, hydrocele, meconium peritonitis (stippled calcification), adrenal haematoma, tumour (yolk sac, teratoma, juvenile granulosa).
- **Prenatal torsion**: hard, non-tender, discoloured at birth; salvage nearly nil → urgent contralateral orchidopexy ± exploration. **Postnatal** (first 30 days): salvage up to 44% → immediate exploration + contralateral fixation. Rationale: prevent asynchronous contralateral torsion.

9 · Landmark trials

Trial	Design	Result	Take-home
PLUTO	RCT, VAS vs conservative in fetal LUTO; 31 randomised, stopped early	Fewer perinatal deaths; no renal benefit; 44% complications	Shunting saves lungs, not kidneys
Great Ormond Street pyeloplasty RCT	APD ≥15 mm, DRF >40%: pyeloplasty vs observation	Most observed stable/improved; minority (APD 20–40 mm) lost function	Observation safe with monitoring
MOMS†	RCT, prenatal vs postnatal myelomeningocele closure	Less shunting, better motor; no urodynamic benefit	Fetal closure not for bladder

35 Urologic Evaluation of the Child

1 · Triage by acuity

Triage asks: **what is lost by waiting** — an organ, a life, or nothing?

Tier	Conditions
Emergent	Acute scrotum; acute abdomen; AKI; obstruction with UTI and/or AKI; retention; incarcerated hernia; paraphimosis; priapism; trauma; newborn ambiguous genitalia, exstrophy, renal vein thrombosis
Urgent (24 h)	Gross haematuria **without infection**; newborn bilateral or solitary-kidney hydronephrosis; renal/testicular tumour; spina bifida
Semiurgent (1–3 wk)	After treated febrile UTI; postnatal dilation (no outlet obstruction); reducible hernia; interlabial mass; amenorrhoea
Routine	Hydrocele; circumcision; incontinence/enuresis; hypospadias; labial adhesions; meatal stenosis; microscopic haematuria; recurrent UTI; undescended testis; varicocele

KEY POINT

Context shifts the tier: hernia emergent if incarcerated, semiurgent if reducible; hydronephrosis urgent if bilateral/solitary.

2 · History

- **Prenatal** (key for hydronephrosis): IVF, growth restriction, maternal diabetes, anomalies, discordant karyotype vs US genitalia; gestation at detection, progression, **amniotic fluid** (proxy for fetal renal function).
- **Consent from guardian, assent from child**; know who holds legal consent (foster care); bullying/abuse may surface. Adolescents: private history.
- Neurogenic bladder: mobility, hand dexterity, sensory level decide self-catheterisation.

3 · Examination

- Minimum systolic BP: **≥60** neonate; ≥70 at 1 mo–1 yr; **70 + 2×age** at 1–10 yr; ≥90 above 10 yr.
- **Shock = tachycardia, tachypnoea; hypotension means decompensation.** Weigh every visit (dosing).
- Male: parent or chaperone; prepubertal supine frog-leg, adolescent standing with Valsalva (hernia, varicocele).
- P1 Testicular development, gonadal descent, abnormalities of descent and bilateral undescended testis — *examining for undescended testis: warm room, sweep from ASIS, cross-legged/squat, ectopic sites; retractile; no imaging*
- **Tanner**: first male sign = testicular/scrotal enlargement (II), not hair; hair to medial thighs = V.
- **Micropenis**: stretched length < mean − 2.5 SD — term **1.9 cm**, adult 9.3 cm; barely changes 5–11 yr; preterm needs gestation-specific values.
- Female: frog-leg; **caudal traction on labia majora** opens the vestibule.
- **Simple sacral dimple** (in gluteal cleft, <5 mm, base visible, <25 mm above anus) → no imaging. Image for lipoma, tag >5 mm, deep/atypical dimple, cleft deviated by a mass.

4 · Urine and blood tests

- Febrile <24 months: **catheter or suprapubic aspirate** for urinalysis and culture; **bag specimens have high false-positive rates** (useful only if negative). †Toilet-trained: clean-catch midstream.
- AAP UTI: pyuria and/or bacteriuria + **≥50,000 colony-forming units (CFU)/mL** (catheter/aspirate) in a symptomatic child.

Test	Threshold
Microscopic haematuria	**>5 red cells (RBC)/high-power field** (adult ≥3), >2 occasions
Calcium/creatinine (hypercalciuria)	>0.4 age 2–5 yr; >0.2 age >5 yr
Protein/creatinine normal	<0.7 first year; <0.4 to 4 yr; <0.2 after 5 yr; >3.0 nephrotic
Specific gravity	<1.010 dilute (water excess, DI); >1.020 dehydration

- Dipstick blood → **confirm by microscopy**. Proteinuria: **first-morning** sample (orthostatic common); persistent on two → nephrology.

5 · Imaging

Only US and MRI avoid radiation (ALARA).

Modality	Answers
US	Kidney, bladder, scrotum, torsion, spine
VCUG	Reflux, urethra, genitogram
CT	Severe trauma, stones
MRI	Tumours, ectopia, duplication, Müllerian/sinus anatomy; urogram gives function
MAG3	**Obstruction** + differential function
DMSA	**Scarring**/pyelonephritis + differential function

6 · Emergent presentations

Acute scrotum

- Commonest: **torsion of appendix testis** (prepubertal; point tenderness; US hyperaemia mimics epididymitis) → supportive.
- **Cord torsion** (mostly >10 yr): severe pain, vomiting, high-riding testis, absent cremasteric reflex (presence does not exclude). **Clinical diagnosis** → explore without imaging; manual detorsion does not obviate exploration; fix contralateral.

TWIST item	Points	Score → action
Testicular swelling	2	0 → no US
Hard testis	2	1–5 → scrotal US
Absent cremasteric reflex; nausea/vomiting; high-riding testis	1 each	**≥6** → explore without US

- **Neonatal (extravaginal) torsion**: firm painless testis, often prenatal; authors prefer urgent exploration.

Other emergencies

- **Ambiguous genitalia**: exclude **salt-wasting CAH** (21-hydroxylase ≈ 95%): crisis day **4–15** → no early discharge; **17-OHP + electrolytes**.

- **Exstrophy**: silk cord tie (no clamp), cling film over plate, transfer.
- **Infected obstruction**: drain promptly (stent/nephrostomy; neonate: pyelostomy, ureterostomy, ureterocele puncture, vesicostomy).
- **Renal vein thrombosis**: haematuria + flank mass + thrombocytopenia; Doppler.
- **Paraphimosis**: exclude **hair tourniquet**; glans compression with traction under analgesia → dorsal slit.
- **Priapism** (≥4 h): ischaemic, usually **sickle cell** → oxygen, fluids, analgesia; cold packs only if sickle cell not suspected. Neonatal priapism benign, resolves in ~4–5 days.
- **Trauma**: haematuria and vitals unreliable → **mechanism** drives CT. Straddle injury → consider **sexual abuse**; meatal blood in a boy → RUG, no blind catheter.
- **Retention**: capacity (mL) = **(age + 2) × 30**. Analgesia, sitz bath, disimpaction before catheter; no cause → neurological evaluation.

7 · Common referral presentations

Antenatal hydronephrosis

- Scan **before discharge** if bilateral, solitary kidney or suspected outlet obstruction (third-trimester APD >7 mm or UTD A2–3); otherwise after **48 h**. Creatinine maternal for 48 h → follow trend.
- Boy with bilateral dilation, thick bladder, dilated posterior urethra → **VCUG before discharge**; catheter meanwhile. Voiding does not exclude obstruction.

Haematuria

- Glomerular: tea-coloured, dysmorphic RBC, casts, protein → C3, C4, ANA, ASO. Urological: bright red, clots, terminal → US; persistent → cystoscopy + retrogrades.
- All gross: urinalysis, culture, US. Haemorrhagic cystitis after marrow transplant (BK) or cyclophosphamide. Wilms commonest renal tumour <15, peak 1–5 years.
- Isolated microhaematuria usually benign (thin membrane, IgA, hypercalciuria) → repeat over 2–3 wk; persistent → US + Ca/Cr.

Bladder bowel dysfunction and enuresis

- Validated questionnaires; **always examine the back**, gait, toe-walking. Behavioural → medical → biofeedback → physiotherapy → psychology; surgery rare.

Scrotal, inguinal and genital

- **Hernia/hydrocele**: communicating hydrocele smaller in morning; parental history/photo suffices; high ligation at internal ring.
- **Undescended testis** with hypospadias or micropenis → suspect DSD.
- **Varicocele** (≈15% adolescents): Lambert volume L×W×H×**0.71**. Non-decompressing supine or right-sided → retroperitoneal US.
- **Testicular tumour**: prepubertal mostly benign (teratoma, epidermoid; malignant = yolk sac) → AFP, β-hCG, LDH guide testis-sparing surgery.
- **Phimosis**: never forcibly retract; lichen sclerosus → treat; steroid if UTI risk.
- **Meatal stenosis**: circumcised boys, deflected thin stream, meatus <5 Fr → meatotomy (EMLA).
- **Labial adhesions** (peak 13–23 mo): observe; symptomatic → oestrogen/steroid cream.

- **Interlabial mass**: US for upstream obstruction; consider rhabdomyosarcoma. **Amenorrhoea** + cyclic pain → outflow obstruction.

> **KEY POINT**
> Red flags: absent cremasteric reflex with acute pain; ambiguous genitalia with impalpable gonads; bilateral hydronephrosis in a boy; fever with obstruction; hypotension; abdominal mass; meatal blood; lumbosacral lipoma or deviated cleft.

8 · Child comfort

- Distraction, midazolam, intranasal ketamine, EMLA avoid general anaesthesia.

36 Pediatric Urogenital Imaging

1 · Principles — ALARA and the three harms

- **ALARA** (ICRP) is the standard; radiosensitivity cannot be predicted, so **every child is treated as radiation sensitive** (developing tissue, long lifetime → stochastic malignancy risk).
- **Sedation/anaesthesia risk usually exceeds contrast risk** (gadolinium NSF in immature neonatal kidneys; iodinated nephropathy).
- MRI: sedation in young children; no fetal MRI in first trimester. US: **safe at all ages and gestations**.

> **KEY POINT**
> Every choice trades **radiation, sedation and contrast** against yield; the long lifespan tips it towards US, and the question is which test fills US's functional gap (obstruction, reflux, pyelonephritis) most cheaply.

2 · Ultrasonography

- P1 USG in neonates — *technique; neonatal pyramids mimic dilation; under-read in first 48 h; SFU grades 0–4; PUV keyhole, MCDK, ARPKD, masses; normal US does not exclude VUR*
- Fetal urine from **10–12 weeks**; by **20 weeks** US detects most anomalies. Check cord insertion and abdominal wall (exstrophy).
- SFU grade 3 heterogeneous; 3→4 defined by parenchyma, not more dilatation. Reporting varies → **review images yourself**.
- US is anatomical. Obstructive vs non-obstructive megaureter/ureterocele → diuretic renography or MRU. **Bilateral hydroureteronephrosis + thick bladder + poor emptying → VCUG** (PUV).

Cysts, masses, stones

Lesion	Side	US hallmark
MCDK	Unilateral	Non-communicating random cysts, scant echogenic parenchyma
ARPKD	Bilateral	Large homogeneously **bright** kidneys, not discrete cysts
ADPKD	Bilateral	Parenchyma almost replaced by cysts
Multilocular cystic nephroma	Unilateral	Like MCDK, more parenchyma; infancy
Cystic Wilms	Unilateral	Overlaps; 2–4 years

- Cysts do not communicate; dilated calyces join the pelvis.
- Neonatal solid mass → **mesoblastic nephroma**; toddler → Wilms; neonate with enlargement + AKI + hypertension → **renal vein thrombosis**. Contrast CT follows US.
- **Stones**: US first and for follow-up. Echogenic **with** shadowing (fat: without); Doppler **twinkling artefact** confirms.

Infection and trauma

- UTI: US not routine at start; **failure to improve on antibiotics → US** (abscess, obstruction). Insensitive for low-grade VUR, duplication, APN, scars (DMSA better). **Normal US neither risk-stratifies APN nor predicts VUR.**

- Trauma: US compares favourably with contrast CT for renal injury follow-up.

Contrast-enhanced voiding urosonography (CeVUS)

- Sulfur hexafluoride microbubbles; five-grade VUR system; **no radiation**. AUC **0.97**, sensitivity ~92%, specificity ~91–94% → recommended for VUR.
- Detects **intrarenal reflux** (voiding, lower pole, rises with grade) — mirrors DMSA scar distribution.

Bladder/urethra

- **Keyhole sign** = thick bladder + dilated posterior urethra → PUV. Pre/post-void volumes.
- Clot vs tumour: scan supine then prone — **clot moves**. Viral cystitis: thick wall mimicking mass, URTI prodrome, gross haematuria.

Scrotum and genitalia

- **Never delay exploration for suspected torsion**.
- **Whirlpool sign**: torsion sensitivity ~96% vs ~76% colour Doppler; specificity ~99%. No flow + heterogeneous parenchyma → non-viable.
- Epididymo-orchitis = hyperaemia (also after spontaneous detorsion); appendage torsion = avascular nodule + surrounding hyperaemia; idiopathic scrotal oedema = **fountain sign**.
- Prepubertal tumours: **teratoma** commonest (solid/cystic/calcified); yolk sac homogeneous vascular; epidermoid **onion rings**, no flow.
- **Do not image routine undescended testis** (AUA Choosing Wisely) except obese child. US primary for internal structures in DSD.

3 · Voiding cystourethrography

- No sedation. Scout may show spinal anomaly. Non-midline catheter → utricle/ectopic ureter.
- **After UTI**: once improving on antibiotics; 4–6-week wait unnecessary. Very early VCUG may **overgrade** VUR (endotoxin ureteric dilatation).
- Indications: hydronephrosis with **dilated ureter(s)**; febrile/recurrent UTI; suspected PUV; boys with epididymitis (ectopic ureter/utricle); UTI in a circumcised boy.
- **Occult VUR**: VCUG finds VUR in only ~40% of DMSA-proven APN. **PIC cystography** (cystoscopic contrast stream at the orifice) for negative VCUG + recurrent febrile UTI.
- **Trauma**: meatal blood → **retrograde urethrogram before any catheter** → gravity fill. Capacity (mL) = **(age + 2) × 30**, max 400. "Pie-in-the-sky" bladder = pelvic fracture, extraperitoneal injury.
- **Genitography** (DSD/cloaca/sinus): defines urethrovaginal confluence (Shopfner classification).

4 · Nuclear medicine

DMSA

- Proximal tubular uptake; APN → photon-deficient areas.

Feature	Acute pyelonephritis	Scar
Contour	Preserved	Volume loss, concavity
Timing	Within days; resolves over ~5 months	Lesion at **≥6 months** = scar

- A third to a half of acute lesions become scars.

MAG3 diuretic renography

- P1 Radionuclide studies and diuretic renography in PUJ obstruction — *MAG3 vs DTPA vs DMSA; well-tempered protocol (hydration, catheter, diuretic timing, ROI); DRF, T½ from maximal filling; biphasic curve; pitfalls*
- Furosemide **1 mg/kg IV** at maximal filling; too early → drainage measured before system full. †DTPA noisier in the immature neonatal kidney.
- **No validated paediatric T½ cut-off** → decide on age, function, T½, curve, upright drainage, serial studies. Progression predictors: **biphasic curve**, poor upright drainage.
- Dietl's crisis: cortical retention up to 48 h after an attack; scan between attacks may be normal.

Radionuclide cystography and testicular scan

- **Direct**: catheter; more sensitive, lower dose than VCUG, poor anatomy, coarse grading → follow-up. **Indirect**: insensitive → not routine.
- Testicular scan: cold = torsion, hot = inflammation; only if equivocal and no delay.

5 · MR urography

- Anatomy + function without radiation: complex anomalies (ectopic ureter, duplication, calyceal diverticulum).
- Protocol: hydration + catheter → **furosemide 1 mg/kg (max 20 mg)** → T2 → gadolinium → dynamic T1; **Patlak** function.
- Limits: up to 1 h; sedation <6 years; equivalence to renography unproven. Poor for stones, infection, trauma, scrotum.
- Best for **bladder rhabdomyosarcoma** (non-enhanced first) and **local staging of renal tumours** + contralateral lesions.

6 · CT and radiography

- CT for speed: trauma, staging; little role in hydronephrosis.
- **Trauma: arterial phase misses collecting-system injury → delayed excretory images mandatory.** CT cystography: intra- vs extraperitoneal (also augment) rupture.
- Stones: best quantification, rarely changes care. KUB: stones, constipation. IVP obsolete. Retrograde pyelography: free retrograde flow does not prove antegrade patency.

7 · Choice of test by clinical question

Question	First line	Next	Avoid
Hydronephrosis — obstruction?	US	MAG3; MRU	CT, IVP
Hydroureteronephrosis ± thick bladder	US	VCUG	—
VUR	VCUG	CeVUS; direct cystogram; PIC if occult	Indirect cystogram
APN / scar	DMSA	—	Grey-scale US; MRI
Stones	US	Low-dose CT; KUB	Reflex CT
Renal mass	US	Contrast CT; MRI	—

Question	First line	Next	Avoid
Bladder mass	US (full bladder)	MRI	—
Trauma	CT with delayed phase; US	RUG; cystography	Catheter before RUG if meatal blood
Acute scrotum	Clinical ± Doppler/whirlpool	Scintigraphy if equivocal	Imaging delaying exploration
Undescended testis	Examination	US only if obese	Routine US, MRI
DSD	US + genitography	MRI	CT

37 Infection and Inflammation of the Paediatric Genitourinary Tract

1 · Overview and the paradigm shift

- UTI affects ~**1% of boys, 3–5% of girls**; 30–50% recur. Commonest bacterial infection in febrile children (~7% of febrile infants).
- Old view: febrile UTI + VUR → scarring, hypertension, CKD → aggressive work-up. **Now**: congenital dysplasia matters more → **selective imaging, prompt treatment, prophylaxis only for the high-risk child**.

2 · Pathogenesis and virulence

- **Ascending**: gut → perineum → bladder → umbrella-cell adherence → invasion → ascent → bacteraemia. Catheter: fibrinogen deposits bind bacteria → biofilm.
- UPEC causes most UTIs; factors let it **adhere, feed, hide, damage**.

Factor	Mechanism	Association
Type 1 fimbriae (mannose-sensitive)	FimH binds uroplakins → entry	Cystitis, ASB
P fimbriae (mannose-resistant)	Bind Galα1–4Galβ (P blood group)	**Pyelonephritis**
Aerobactin	Siderophore	Growth in iron-poor urine
α-Haemolysin, CNF1	Cell lysis, cytoskeletal damage	Inflammation, invasion
Capsule (K antigen)	Antiphagocytic, anticomplementary	Pyelonephritis
Biofilm, flagella	Shield; motility	Recurrence; ascent

- Host: TLR4 → cytokines → neutrophils. No attachment → asymptomatic carriage; failed clearance → destructive inflammation.
- **Intracellular bacterial communities** → **quiescent intracellular reservoir** for months → reseed lumen → same-strain recurrence despite adequate antibiotics.

> **KEY POINT**
> IBCs and biofilms shield bacteria from neutrophils and antibiotics; a biofilm survives 1000–1500× normal killing concentrations (poor penetration, slow growth, altered targets).

3 · Host risk factors

- First year: boys **3.7%** vs girls 2%; by 6 years boys 2%, girls 7%. Febrile girl 2 months–2 years ≈ 2.3× febrile boy's risk. White > Black.
- P1 phenotype in girls with recurrent pyelonephritis; Lewis Le(a−b−); non-secretors → scarring.
- **Circumcision**: UTI in first 6 months ↓ **~10-fold**. US Army cohort: circumcised 0.11%, uncircumcised 1.12%, girls 0.57% MCQ. Number needed to circumcise: **111** normal boys, 11 recurrent UTI, **4** high-grade VUR. Protective in antenatal hydronephrosis.
- **Faecal reservoir**: same strain cycles; antibiotics (esp. TMP-SMX) destroy protective flora.
- **VUR**: 1–2% of newborns, **25–40%** after first UTI; most pyelonephritis occurs without VUR, but grade III–IV doubles pyelonephritic change → *Ch 41*.

- **BBD** (infrequent voiding, incomplete emptying, constipation, overactivity): treating it cuts recurrence, aids VUR resolution.
- **Neurogenic bladder**: ~2.5 UTIs/year; on CIC 40–80% bacteriuric — **ASB without VUR needs no antibiotics**; prophylaxis breeds resistance; catheter type irrelevant; frequent catheterisation prevents UTI.
- **CAUTI**: commonest nosocomial infection; 3–10% per catheter-day → remove early.
- IgG nadir 1–3 months, little urothelial IgA → highest risk early infancy. Obesity raises risk.

> **KEY POINT**
> Host factors either raise inoculum/dwell time (prepuce, faecal flora, stasis, constipation, catheter) or lower defence (immature immunity, receptor-bearing blood groups); prevention reverses whichever applies.

4 · Classification

- **Complicated** (anomaly, dysfunction, foreign body; all neonatal/infant UTI) vs uncomplicated; upper vs lower; first vs recurrent.
- Febrile UTI (presumed pyelonephritis) vs afebrile cystitis. Localisation poor: only **50–66%** of febrile children have acute DMSA changes (gold standard).

Recurrence	Between episodes	Cause	Remedy
Unresolved bacteriuria	Never sterile	Non-compliance, malabsorption, resistance	Culture-directed retreatment
Bacterial persistence	Sterile; same organism	Nidus: stone, anomaly, papilla, catheter, stent	**Remove nidus**
Reinfection	Sterile; new organism	New ascent	Prevention

Asymptomatic bacteriuria

- Kass: two cultures **≥10^5 colony-forming units (CFU)/mL**, same organism, no symptoms.
- Infant (<1% term, 3% preterm): treat + RBUS. School-age girls without VUR/anomaly: **do not screen or treat** (selects resistance, replaces harmless strain).

5 · Diagnosis

Clinical

- Fever ≥38 °C. Bacteraemia predicted by toxic look, age <30 days, ≥39.4 °C. Infants nonspecific (irritability, poor feeding, jaundice, vomiting).
- <2 years: fever >40 °C, previous UTI, suprapubic tenderness, **uncircumcised penis**. **Another fever source (otitis) only halves UTI risk.**
- Examine genitalia (phimosis, labial adhesions, ectopic ureter), spine, constipation. Adolescents: gonorrhoea, Chlamydia.

Urine collection

Method	Contamination	Role
Bag	50–60% (false-positive culture up to 75%)	**Reliable only when negative**; never culture
Clean catch	~26%	Toilet-trained; NICE preferred

Method	Contamination	Role
Catheter	10–15%	Sens 95%, spec 99%; **standard <2 years non-toilet-trained**
SPA	<1%	Reference; US guidance → 80–90% success

- Collect **before antibiotics** unless delay endangers (NICE). AAP 2–24 months: bag urinalysis → if positive, catheter/SPA culture. Examine within 1 h or 4 h refrigerated.

Test	Sensitivity	Specificity	Use
Leukocyte esterase (LE)	~79%	~87%	Negative helps **rule out**
Nitrite	~50%	**~98%**	Positive **rules in**
LE or nitrite or microscopy	99.8%	70%	Best rule-out
Pyuria ≥5 WBC/HPF	~74%	~86%	No advantage over LE
Gram stain, fresh urine	~91%	92–96%	Most reliable rapid test

- **Nitrite false negatives**: short bladder dwell (infants), dilute urine, gram-positives (Enterococcus).
- Pyuria less with Klebsiella/Enterococcus → culture anyway; UTI without pyuria rare (separates ASB). Threshold **≥10 WBC/HPF** better <2 years.
- **Culture essential**; unnecessary if urinalysis entirely normal.

Guideline	Specimen	Significant count (CFU/mL)
Kass	Voided	$\geq 10^5$
AAP 2011 (2–24 months)	Catheter/SPA	**≥50,000 + pyuria**
EAU	Midstream/catheter	10^3–10^4 monoculture
RCH Melbourne	SPA / catheter	Any gram-negative / $>10^3$
ISPN (Italy)	Catheter / midstream	$>10^4$ / $>10^5$

- **The cleaner the specimen, the lower the accepted count.** Contamination: low counts, mixed growth, lactobacilli, CoNS, diphtheroids.
- CRP sensitive, not specific; **procalcitonin ≥0.5 ng/mL** best predicts APN and later scarring.

6 · Imaging after UTI

- No evidence routine imaging prevents scarring, hypertension or renal failure → **order a test only if it changes management**.
- US: abnormal in ~15% after first febrile UTI, only 1–2% need action; **urgent if unwell or not responding at 48 h**. Pitfalls: oedema, lobar nephronia mimics tumour, endotoxin dilates pelvis. Poor for VUR/scars.
- VCUG: gold standard for VUR; sterile urine, well child; cyclic filling ↑ sensitivity; volume at reflux onset predicts resolution. CeVUS alternative.
- DMSA (~1 mSv): APN best within **1 week**; scar at **≥6 months**; ~15% of acute lesions scar; scar indistinguishable from dysplasia. CT (10–15 mSv) only for occult stones or mass vs inflammation.

Guideline	Ultrasound	VCUG	Late DMSA
AAP (2–24 months)	Yes, after first febrile UTI	If US abnormal (or recurrence)	No

Guideline	Ultrasound	VCUG	Late DMSA
NICE <6 months	Yes	Atypical/recurrent	Atypical/recurrent
NICE >6 months	Atypical/recurrent	No	Atypical/recurrent
Top-down	No	If acute DMSA abnormal	If acute DMSA abnormal
RCH Melbourne	If <3 months or seriously ill	No	No

- **NICE "atypical"**: seriously ill, poor urine flow, abdominal/bladder mass, raised creatinine, septicaemia, no response within **48 h**, non-*E. coli*. Late DMSA at 4–6 months.
- NICE/AAP miss ~half of grade III–V VUR but spare most children VCUG; high-grade VUR declares itself at recurrence. AAP Urology Section still accepts VCUG (RIVUR).

7 · Treatment

- Urine sterile within 24 h of effective therapy. **Delay matters**: new-scar odds double if antibiotics start after 72 h vs within 48 h. Start empirically after urine collected.
- **Oral = IV** for non-toxic children >2 months tolerating oral drugs.
- **Admit**: <1 month (some <2–6 months), toxic, dehydrated, vomiting, doubtful compliance. Neonates: septic work-up (bacteraemia ~20%, hyponatraemia, hyperkalaemia) → IV → oral after ≥24 h afebrile; ≤2 days IV safe in young infants.
- 90% afebrile by 48 h; **not improving at 48 h → RBUS + broaden cover**.

Infection	Duration
Febrile UTI	**7–14 days** (shorter inferior)
Afebrile cystitis	**2–4 days** (beats single dose; equals 7–14 days)
Lobar nephronia	≥3 weeks
Renal abscess	Antibiotics; drain if no response
Pyonephrosis	Antibiotics + drainage

- Choose by **local antibiogram**. TMP-SMX and amoxicillin poor empirical agents. Prefer **cephalexin**; nitrofurantoin for cystitis. Urinary levels >100× serum breakpoints → don't escalate needlessly.
- *E. coli* 83% in outpatient girls, ~37% in inpatient boys; **Proteus is a boys' organism**.
- **Neonates**: cover Enterococcus → **ampicillin + third-generation cephalosporin or aminoglycoside**.
- Broad cover after previous UTI, recent antibiotics/admission, anomaly; aminoglycoside for Pseudomonas. Fluoroquinolones reserved (FDA: complicated UTI/pyelonephritis).

Agent	Dose (mg/kg/day)	Restriction
Cephalexin	50–100 in 4	—
Cefixime	8 once daily	—
Amoxicillin–clavulanate	20–40 in 3	—
TMP-SMX	6–12 (TMP) in 2	**<6 weeks/premature**: kernicterus
Nitrofurantoin	3–5 in 2	**<3 months**, GFR <50%, G6PD deficiency; **never for febrile UTI**
Ceftriaxone (IV)	75 once daily	—

Agent	Dose (mg/kg/day)	Restriction
Gentamicin (IV)	7.5 (once daily acceptable)	Nephro-/ototoxicity †

No repeat culture if organism was sensitive.

8 · Renal scarring and sequelae

- Scars favour the **poles** (compound papillae → intrarenal reflux); risk rises with VUR grade.

Pyelonephritic episode	1st	2nd	3rd	4th	5th
Scar risk	5%	10%	20%	40%	60%

- Small kidney, diffusely low uptake = usually **dysplasia**. Steroids not routine.
- Childhood UTI = **<1% of CKD**; ESKD ~1 in 10,000. Scars → hypertension, pre-eclampsia.
- Follow height, weight, BP; bilateral scars → lifelong BP, function, proteinuria (RAS blockade).

9 · Prevention and prophylaxis

- Recurrence **10–30%**, highest in first 3–6 months; girls up to 40%.
- **CAP**: not after first UTI (AAP, NICE). For **high risk**: girls with dilating VUR (≥III), BBD, recurrent febrile UTI. Low risk: circumcised boy, no BBD, normal imaging.
- Dose **~¼ therapeutic** at bedtime; differ from treatment agent (prior exposure triples resistance). TMP-SMX (not <6 weeks), nitrofurantoin (spares faecal flora), cephalosporin (young infants).
- **BBD**: VUR + BBD → recurrence ~45% vs 15%. Timed voiding + laxatives → anticholinergics → biofeedback → CIC *→ Ch 47*.
- Non-antibiotic: **circumcision** for high-risk boys (recurrent febrile UTI, high-grade VUR, hydroureteronephrosis, obstructive uropathy); topical steroid for phimosis reasonable; cranberry mixed, probiotics adjunct, D-mannose unproven.

10 · Special infections

- P1 Xanthogranulomatous pyelonephritis. / Fungal UTI — *XGP: Proteus, stone–obstruction triad, mimics tumour, nephrectomy (diffuse) vs partial (focal), hard dissection; Candida risks, C. glabrata resistance, treat >10,000–15,000 CFU/mL, fungus ball drainage + amphotericin*
- **Lobar nephronia**: CT wedge hypoenhancement mimicking mass → antibiotics ≥3 weeks. **Pyonephrosis → drain**. **Renal abscess**: urine culture negative in up to 20%; US detects ≥1 cm, CT most accurate.
- Paediatric XGP: calculi 38–70%; failure to thrive, anaemia.
- **Viral haemorrhagic cystitis**: adenovirus 11 (and 21); after marrow transplant severe disease 50–80% mortality. **BK virus → cidofovir**.
- †**Prepubertal epididymitis**: usually sterile; positive culture/recurrent → US + VCUG for ectopic ureter, utricle, obstruction. Exclude torsion.
- †**Balanoposthitis**: soaks, hygiene, topical agent; recurrent or pathological phimosis → circumcision.

11 · Landmark trials

Trial	Design	Result	Take-home
RIVUR	607 children, VUR I–IV, TMP-SMX vs placebo, 2 years	Recurrent UTI ~halved (NNT 8); **no difference in scarring**	Prevents recurrence, not scars; most benefit with BBD, febrile index UTI
Swedish Reflux Trial	Randomised, dilating VUR, 1–2 years	Prophylaxis reduced recurrence in girls	CAP for girls with dilating VUR

38 Core Principles of Perioperative Management in Children

1 · Perioperative surgical home and enhanced recovery

- **Perioperative surgical home** (ASA): surgeon–anaesthesiologist-led standardised pathways to post-discharge.
- **ERAS**: counselling, shorter fasting, judicious IV fluid, no routine tubes/drains, opioid-sparing analgesia, early feeding/mobilisation → shorter stay, no more complications.

2 · Preoperative assessment

- Risk raised by URTI, asthma exacerbation, dehydration, uncontrolled diabetes, prematurity, fever.
- **Preoperative anxiety** (up to half) → more pain, emergence delirium, new behaviour problems (enuresis) → child life specialists, explanation, play.
- **Jehovah's Witness**: child not allowed to die for lack of blood (court order); correct anaemia/coagulopathy beforehand.
- **Investigations indication-driven**: no routine bloods, CXR or ECG; murmur alone is not an ECG indication. Urine hCG in females of menstrual age — MCQ false positive after augmentation cystoplasty.
- **Urine culture** before any procedure opening the urinary tract; reschedule until sterile. Always culture in neurogenic bladder; treat asymptomatic bacteriuria before endourology (mucosal injury → sepsis).
- **URTI**: airway hyperreactive **6–8 weeks** → laryngospasm, bronchospasm, desaturation; laryngeal mask safer than endotracheal tube. Postpone elective surgery ≥ 4, ideally 6 weeks.
- **Sepsis**: antibiotics within **1 h** of septic shock; avoid neuraxial blocks. **Trauma**: full stomach → rapid-sequence induction.

3 · Fasting (NPO)

Minimum fast	Material
2 h	Clear liquids, carbohydrate drinks (European and Canadian 2022: 1 h)
4 h	Breast milk
6 h	Infant formula, non-human milk, light meal, tube feeds
8 h	Regular meal

- Trap: non-human milk = light meal. Youngest first (dehydrate faster); inpatients on IV maintenance; medicines with clear fluid to 2 h.
- **Metabolic**: ketogenic diet — no carbohydrate (incl. suspensions); glycogen storage/urea cycle disorders — glucose fluid throughout; mitochondrial disease — avoid lactated Ringer and propofol.
- **Diabetes**: first case; metformin stopped 24 h before; insulin adjusted, not stopped (half long-acting dose night before); long case/emergency → IV insulin.
- Omit ACE inhibitors that morning; stop warfarin/aspirin 7–10 days before; never stop clonidine abruptly (rebound hypertension).

4 · Anaesthesia considerations

Neonate and preterm infant

- Little reserve: surfactant deficiency, BPD, small blood volume, retinopathy of prematurity (accept saturation low 90s), hypothermia.
- **Postoperative apnoea** (peaks 12–23 h) → 23-h cardiorespiratory monitoring if postconceptional age **< 50–60 weeks** (ex-preterm) or **< 44 weeks** (term); defer elective surgery beyond these.

Anaesthetic neurotoxicity

- FDA warning (2016): repeated or lengthy (**> 3 h**) anaesthesia at **< 3 years** or third-trimester pregnancy may affect brain development — based on animal neuroapoptosis.
- Human trials of single short exposure negative (§11). Counsel: single exposure < 1 h not shown harmful; do not delay necessary surgery.

Comorbidities

- **Asthma**: pre-induction β_2-agonist; stress steroids if on systemic/daily inhaled steroids.

  Congenital heart disease: highest risk — single ventricle, pulmonary hypertension, LV outflow obstruction, cardiomyopathy.
- **CKD**: anaemia, platelet dysfunction, hyperkalaemia; avoid surgery within 4–6 h of haemodialysis; no NSAIDs in CKD 3–4.
- **Sickle cell**: avoid dehydration (most important: first case, IV fluid during fast), hypoxia, acidosis, cold, stasis (no tourniquets). Acute chest syndrome is the main cause of death.
- **Phaeochromocytoma**: α-blockade *before* β-blockade + volume loading; post-resection hypoglycaemia.
- **CAH**: IV stress-dose hydrocortisone with glucose-containing fluid.

Spina bifida

- Confirm VP shunt function (malfunction → postpone); scoliosis → restrictive lungs.
- **Latex allergy** (repeated mucosal exposure) → anaphylaxis; †latex-free from birth.
- **Autonomic dysreflexia** (lesion ≥ **T6**): bladder distension/noxious stimulus → severe hypertension, bradycardia, stroke. † Stop stimulus, drain bladder, sit up, rapid antihypertensive.

5 · Fluid and electrolyte management

- **Holliday–Segar 4-2-1**: **4 mL/kg/h** first 10 kg + **2 mL/kg/h** next 10 kg + **1 mL/kg/h** per kg above 20. 22 kg → 40 + 20 + 2 = 62 mL/h.
- Large-volume saline → hyperchloraemic acidosis. Dextrose under 2–4 months; no potassium if dialysed.

Maintenance fluid	Historical	Current
Tonicity	Hypotonic (0.45% saline)	Isotonic (0.9% saline or lactated Ringer) with dextrose
Potassium	10 mEq/L if < 10 kg; 20 mEq/L if > 10 kg	

KEY POINT

Why isotonic? Surgical stress, pain, nausea, hypoxia → non-osmotic ADH release → aquaporin-2 water retention. Hypotonic fluid adds free water → hyponatraemia (< 135 mmol/L). † Large brain-to-skull ratio → little room for cerebral oedema; encephalopathy can be fatal.

6 · Blood loss and transfusion †

- EBV: preterm 90–100, term neonate 80–90, infant 75–80, child 70–75 mL/kg.
- Maximum allowable blood loss = EBV × (starting − target haematocrit) / starting haematocrit; crossmatch when expected loss approaches it.
- Transfuse at haemoglobin **< 7 g/dL** in stable children (higher in neonates, cyanotic heart disease, active bleeding); 10–15 mL/kg raises haemoglobin ≈ 2–3 g/dL.

7 · Temperature

- Infants lose heat fast (surface area, little brown fat) → acidosis, coagulopathy, apnoea. Warm theatre, †forced-air warming, warmed fluids, core temperature; laparoscopy adds gas cooling → *Ch 39*.

8 · Antibiotic prophylaxis and intraoperative crises

- **Timing**: within **60 min** of incision; vancomycin and fluoroquinolones within **120 min**.
- Doses: cefazolin **30 mg/kg** (redose 4 h); ampicillin 50 mg/kg (2 h); gentamicin 2.5 mg/kg, vancomycin 15 mg/kg, metronidazole 15 mg/kg, ciprofloxacin 10 mg/kg (not redosed).

	Anaphylaxis	Malignant hyperthermia (MH)
Triggers	Neuromuscular blockers, β-lactams, latex	Succinylcholine, volatile agents (not nitrous oxide)
First sign	Collapse, bronchospasm, hives minutes after induction	Unexplained hypercarbia, tachycardia, masseter spasm; late — acidosis, hyperkalaemia, rigidity
Treatment	**Adrenaline**, 100% oxygen, fluids; tryptase	Stop triggers, **dantrolene**, clean circuit; admit 24–48 h (recrudescence)

9 · Analgesia

- Self-report impossible < 3 years → FLACC (Face, Legs, Activity, Cry, Consolability).
- **Regional blocks under GA**: safe, opioid-sparing — penile, **caudal**, spinal, epidural (from 2 kg). Ultrasound-guided pudendal block for hypospadias: less pain and opioid than caudal. † Caudal 0.25% bupivacaine 0.5 mL/kg (sacral) to 1 mL/kg (lower thoracic); maximum bupivacaine 2.5 mg/kg.
- **Multimodal, scheduled**: acetaminophen 15 mg/kg 6-hourly + ibuprofen 10 mg/kg 8-hourly; ketorolac 0.5 mg/kg IV 8-hourly ≤ 72–96 h (renal toxicity); no NSAIDs < 6 months. Adjuncts: IV lidocaine, ketamine, dexmedetomidine.
- Opioids for breakthrough; PCA from ≈ 4 years. **Codeine contraindicated**: poor metabolisers no analgesia, ultrarapid metabolisers lethal morphine. Methadone needs ECG (QT).
- Discharge opioids: 3–5 days' supply; teach safe disposal.

10 · Postoperative care, VTE and day-case discharge

- Laryngospasm → positive pressure → propofol → reintubation. Bronchospasm → salbutamol → adrenaline, steroids, magnesium.
- **Emergence delirium** (2–8 years, within 30 min): does not recognise parents (child in pain does). Low-dose propofol, dexmedetomidine or fentanyl.
- **VTE** †: rare before puberty; at-risk adolescents (immobility, obesity, oestrogen, CVC, malignancy, pelvic surgery, prior VTE) → mobilisation, mechanical prophylaxis; LMWH if several factors.
- **Day-case**: †ASA I–II, outside apnoea-risk ages, short low-blood-loss procedure. Discharge criteria: Aldrete or Post-Anesthetic Discharge Scoring System.

11 · Landmark trials

Trial	Design	Result	Take-home
GAS	Randomised; infant hernia, awake regional vs sevoflurane < 1 h	No cognitive difference at 2 and 5 years	Single short anaesthetic safe
PANDA	Sibling-matched; single exposure < 3 years	No IQ difference at 8–15 years	Repeated exposure untested
MASK	Cohort; exposure < 3 years	Negligible deficit after single exposure	Reassuring

39 Principles of Laparoscopic and Robotic Surgery in Children

1 · Evolution and indications

- P3 Discuss diagnostics and operative pediatric laparoscopy in urology — *evolution 1976 → 1994 → 1995 → 2002; indications; age IAP limits; only contraindication; access; complications (right common iliac, inferior epigastric, port-site hernia 0.5–3%)*
- First paediatric urological laparoscopy: non-palpable testis, **1976**. Intracorporeal suturing was the bottleneck; the wristed robot removed it → growth in **reconstruction** (pyeloplasty commonest; reimplantation, (hemi)nephrectomy, intra-abdominal testis, urachal/diverticulum excision; deep pelvis — utricle, urogenital sinus, bladder neck).
- Only strict contraindication: **inability to tolerate pneumoperitoneum**. Age and size are not contraindications; cardiopulmonary disease needs optimisation.

2 · Physiology of pneumoperitoneum

- P1 Pneumoperitoneum physiology; Laparoscopic access and gases — *pressure vs CO_2 effects on heart, lungs, kidney, ICP; vagal bradycardia; CO_2 choice; gas embolism signs and treatment (left lateral head-down, oxygen, aspirate)*

Effects are pressure-dependent and reverse on desufflation → **lowest age-appropriate pressure**: IAP < 12 mm Hg; ≈ 7 mm Hg feasible.

Age	0–1 month	2–12 months	1–2 years	2–8 years
Maximum IAP	**≤ 6**	**≤ 8**	**≤ 10**	**≤ 12 mm Hg**

- **Cardiovascular**: IAP > 10 mm Hg ↓ venous return and cardiac index; catecholamines and RAAS ↑ systemic vascular resistance. Insufflation **vagal reflex** → warn anaesthetist, desufflate, reinsufflate slowly.
- **Respiratory**: infants have high chest wall compliance, low FRC → ↓ tidal volume, ↑ peak pressure; worse with Trendelenburg. CO_2 absorption affects infants most.
- **VP shunts**: feared raised ICP, but MIS generally safe.
- **Renal**: transient oliguria, most marked < 1 year, reversible → **no fluid boluses**.
- **Hypothermia** †: cold dry gas cools infants → warmed humidified gas, low flows, minimise leaks.
- **CO_2 embolism** †: falling end-tidal CO_2, desaturation, collapse, "mill-wheel" murmur.

3 · Access

- Open (Hasson) vs closed (Veress: aspirate → saline drop test → low opening pressure): no difference in entry failure or vascular/visceral injury (Cochrane). What protects is confirming intraperitoneal position, not the method.
- Veress at umbilicus; after scarring/failure, **Palmer's point** — left midclavicular line ≈ 3 fingerbreadths below costal margin. Preperitoneal tip → high opening pressure → subcutaneous emphysema.

4 · Ports, instruments and placement

- **3- and 5-mm** laparoscopic ports suit infants; robots need **8 mm**. Valveless trocars: constant pressure, less CO_2, but subcutaneous emphysema.
- Laparoscopic instruments 4 degrees of freedom (DOF); robotic wristed 7 DOF.
- Energy: monopolar insulation failures → remote burns; bipolar/ultrasonic seal vessels up to 7 mm.
- Triangulation essential in laparoscopy; robot tolerates linear placement. **HIdES** (hidden incision endoscopic surgery): ports below the Pfannenstiel line.
- **Positioning**: renal — modified flank at bed edge, 30–45°, padded, test table rotation; pelvic — supine (frog-leg), steep Trendelenburg, strapped, face protected.

5 · Approaches

	Transperitoneal	Retroperitoneal	Vesicoscopic
Space	Largest; familiar	Limited	Limited; small bladder predicts complications
Strength	Complex reconstruction	Avoids peritoneum; preserves peritoneal dialysis	Intravesical access
Limitation	Colon mobilised or mesenteric window	Unfamiliar view; peritoneal tear collapses space	Learning curve; port-site closure
Uses	Testis, kidney, ureter, bladder	Pyeloplasty, (hemi)nephrectomy, lymph node dissection	Cross-trigonal reimplantation

- **Left pyeloplasty**: transmesenteric window avoids colonic mobilisation.
- **Retroperitoneoscopy** †: open entry below 12th rib tip, finger/balloon dissection, flank or prone. Transperitoneal and retroperitoneal nephrectomy outcomes equivalent; retroperitoneal for peritoneal dialysis patients.
- **Vesicoscopy**: camera through bladder dome under cystoscopic view, urethra occluded, CO_2 insufflation; robotic version limited by 8-mm ports.

6 · Robotics in children

- **Working space** ≈ **1 L at 1 year vs 6 L in adults** → collisions. Aids: wide port spacing, linear placement, "burping" ports, minimal inhalational agent, enema and rectal tube.
- Infant robotic pyeloplasty safe; vs open: shorter stay, less narcotic.
- **Limitations**: no 5-mm robotic ports; single-port needs a 25-mm cannula and ≈ **10 cm** to target — prohibitive in small children; no haptics; cost; training.
- **Benefits**: shorter stay (hours for pyeloplasty, days for Mitrofanoff/augmentation); cosmesis — incisions < 15 mm contract, larger enlarge; pain benefit least proven.

7 · Complications

- Major vascular: **right common iliac** (behind umbilicus, Veress); minor: **inferior epigastric** (ports under vision). Decompress bladder/stomach †; inspect on exit.
- Port-site hernia highest around 1 year → **close fascia** at every port.
- Experience strongest predictor (> 12 cases/year → fewer). Pyeloplasty complications equal open/laparoscopic/robotic. Same-day discharge feasible after uncomplicated robotic

renal/ureteric surgery without drains.

KEY POINT

Ergonomics: laparoscopy strains hands, thumb, shoulders → table at 70–80% elbow height, monitor at eye level. Robot offloads the arm but fixes the neck; use the clutch to keep elbows on the armrest.

40 Clinical and Urodynamic Evaluation of Lower Urinary Tract Dysfunction in Children

1 · Principles and terminology

- Diagnosis mostly by history, examination, non-invasive tests: exclude anatomical/neurological cause → define **storage vs emptying** dysfunction (ICCS 2014).
- **Expected bladder capacity (EBC)**: infant ≤1 y = **weight (kg) × 7 mL**; older = **(age in years + 2) × 30 mL**. Small <65% EBC; large >150% (some >130%). Functional capacity (largest diary void, or void + PVR) normal ≈70–115% EBC.

2 · History

- **Secondary enuresis** = wetting after **≥6 months** dry → look for stressors.
- Girls — **toilet posture** (legs together, perching, dangling feet) activates pelvic floor → high voiding pressure, incomplete emptying, UTI. Fix: footstool or sit backwards.
- **Holding manoeuvres** (potty dance, squatting, perineal grabbing) suppress DO via the sacral reflex arc — same mechanism as neuromodulation.
- **Bowel**: shared S2–S4; may be the *sole* cause of urinary symptoms. Target **Bristol 3–4 daily**, painless.
- **Rome IV functional constipation** (developmental age ≥4 y, ≥1 month): **≥2 of** — ≤2 stools/week; ≥1 soiling/week; retentive posturing; painful/hard stools; large rectal mass; stools blocking the toilet.
- Bagged urine reliable only if negative; febrile culture-proven UTI suggests stasis. **Neuro red flags**: perineal numbness, gait/limb change with new LUTS.

> **ALGORITHM**
> BBD: Rome IV → 7-day diary + urinalysis + pelvic US → **treat bowel first** → LUTS persist → uroflow + EMG → urotherapy → anticholinergic → β3-agonist → botulinum toxin / biofeedback / neuromodulation.

- **7-day bowel–bladder diary**: voids, stools, Bristol type, wetting — removes recall bias. **48-h frequency–volume chart** (2 × 24 h, need not be consecutive): capacity vs EBC, polyuria; **mandatory** when first-line therapy fails.
- Scores: **DVSS** best tracks treatment; **DVISS** best for diagnosis; PinQ measures burden, not response.
- Psychological comorbidity highest with **faecal incontinence** (> daytime > enuresis).

3 · Physical examination

- **Every visit**: height, weight, BP.
- **Neurological**: include perineal sensation, anal tone. Hand function decides self-catheterisation. Functional level and ambulation **do not correlate**.
- **Sacral reflexes**: anal wink and bulbocavernosus → external anal sphincter contracts; absent = interrupted S2–S4 arc.
- **Spine**: simple sacral dimple (<2.5 cm from anus) low risk → spinal US up to 6 months. High risk (hair tuft, lipoma, vascular malformation, multiple stigmata, VACTERL) → **MRI**. US useful only until arches ossify (≈10–12 months).

- **Genitalia**: meatal stenosis matters only with stream deviation; labial fusion (topical oestrogen); **continuous incontinence in a girl → ectopic ureteric orifice**.

4 · Laboratory tests

- **Urinalysis for every child with LUTS**; negative reassures. **Culture not routine**; on CIC culture only with fever, pain, malaise, new/worse incontinence or spasm. Bloods only if renal disease suspected.

5 · Non-invasive urodynamics

- P3 P1 Uroflowmetry; EMG — *bell/tower/staccato/interrupted/plateau patterns and meaning; Qmax peak ≥2 s; patch EMG at 3 and 9 o'clock; lag time <2 s DO, 2–6 s normal, >6 s bladder-neck dysfunction*
- Uroflow valid only if toilet-trained, voided volume **≥50% EBC** (pattern needs ≈50–100 mL), >1 curve; emptying data only.
- Staccato never reaches zero; interrupted does. Pelvic floor should be **silent during voiding**.

> **KEY POINT**
> Curve = detrusor vs outlet. Too much detrusor → tower; too much outlet → plateau; intermittent outlet activity → staccato; intermittent detrusor effort → interrupted. Patch EMG tells which side is at fault.

- **Pelvic US**: bladder wall abnormal **>3 mm full / >5 mm empty**; rectal diameter **>30 mm** without urge = faecal loading.
- **PVR** (repeat — variable): abnormal at 4–6 y **>30 mL or >21%** of capacity; 7–12 y **>20 mL or >15%**; repeated >20 mL abnormal. Valid only if **<5 min** after void with bladder 50–115% EBC — late or over-distended = invalid, not abnormal.

6 · Invasive urodynamics

Indications

- **Neurogenic**: spinal dysraphism, tethered cord, sacral agenesis, sacrococcygeal teratoma, transverse myelitis, SCI, suspected neurogenic bladder with new LUTS.
- **Severe obstruction**: PUV, urethral atresia, large ectopic ureterocele.
- Non-neurogenic LUTS: only after failed 2nd/3rd-line therapy — study only if it will change management.

Technique

- **Avoid sedation/GA** (alters detrusor); if unavoidable, ketamine over midazolam.
- Routine culture/prophylaxis not needed — post-study UTI ≈ **1.4%**
- **6 Fr dual-lumen** urethral + rectal catheter; perianal patches; **fill 5–10% of EBC per minute**, warm infusate.
- **Stop filling** at strong urge, discomfort, voiding, leak > fill rate, Pdet persistently **>40 cmH_2O**, or volume **>150% EBC**.
- First fill gives highest Pdet — report it (kidney-protective). Natural-fill: smaller capacity, higher voiding pressure.
- **Videourodynamics** adds trabeculation, diverticula, VUR and the volume/pressure at onset, bladder-neck shape.

Storage phase

- **Pdet = Pves − Pabd** (zeroed at symphysis); normal filling flat. **DO**: phasic involuntary rises (exclude cough/strain). **Underactivity**: large, acontractile, voids by straining.
- **Compliance = ΔV/ΔPdet**, calculate every study: **>20 normal; 10–20 moderately; <10 severely impaired**. In small contracted bladders absolute Pdet (<30–40) is the better guide; an **early** rise means most of the day at high pressure.
- P1 Leak point pressures (VLPP, ALPP, DLPP) in SUI: clinical implications — *DLPP = outlet resistance, high = upper-tract risk; ALPP = sphincter competence, low = bad; McGuire 1981*
- DLPP threshold **>40 cmH₂O** (or no leak at high pressure) = danger; ALPP has no paediatric norms.

> **KEY POINT**
> **McGuire (1981)**, myelodysplasia: leak pressure >40 cmH₂O → hydronephrosis or secondary VUR in **85%**; pressure is transmitted up the ureters. Aim for the lowest achievable pressure.

Voiding phase

- Valid if ≥50–100 mL and ≥50% functional capacity voided. **High Pdet + low Qmax = obstruction**: anatomical — plateau, **quiet EMG**; functional (DSD/dysfunctional voiding) — staccato/interrupted, **active EMG**.
- Normal Pdet at Qmax ≈ boys 66, girls 57 cmH₂O; infant boys up to 118 (physiological) MCQ.

> **KEY POINT**
> **Upper-tract threats**: compliance <20 (esp. <10); DLPP >40 cmH₂O; sustained high filling Pdet; DSD with poor emptying; high voiding pressure with poor flow. Interobserver agreement poor.

- **UMPIRE** (spina bifida): urodynamics ≈**3 months** → 1, 2, 3 years → yearly to 5 y if VUR or hostile bladder. Hydronephrosis is late and insensitive; re-study on any change and before reconstruction.

7 · Evaluation in specific conditions

Condition	Typical findings	When to study
Non-neurogenic LUTD / enuresis	DO, dysfunctional voiding	Only if refractory
Spinal dysraphism	UMPIRE risk by DLPP/end-fill Pdet: **<25 low; 25–39 or DO intermediate; ≥40 high**	Baseline, serial, pre-surgery
Sacral agenesis	Areflexia + weak sphincter, or DO + DSD; level does not predict; ⅓ diagnosed late (flat buttocks, short cleft)	If voiding dysfunction
Anorectal malformation	Tethered cord common	Baseline ≈1 y after repair; on change
PUV	DO → poor compliance → myogenic failure; polyuria worsens ("valve bladder" ≈15%)	Serial
Cerebral palsy	UMN; DO, large or small capacity, DSD rare	Only if conservative therapy fails
SCI	Spinal shock → retention; higher lesions → more DSD	Video-urodynamics 3–6 months after injury

Condition	Typical findings	When to study
Transverse myelitis	DO > poor compliance > DSD > DLPP >40; motor recovery does not predict bladder	Early, with CIC

KEY POINT

UMPIRE dropped DSD from high risk because reviewers could not agree who had it — not proof DSD is benign. Neurogenic bladder goals: normal capacity, low storage pressure, complete emptying, upper-tract protection.

8 · Landmark trials

Trial	Design	Result	Take-home
McGuire 1981	Myelodysplasia urodynamic cohort	DLPP >40 cmH_2O → upper-tract damage in 85%	40 cmH_2O = safety threshold

42 Bladder Anomalies in Children

1 · Classification and embryology

- P3 Embryological development of urinary bladder and management of Hutch Diverticulum? — *cloacal division weeks 4–6, UGS fates, trigone, urachus months 4–5; anomaly table; Hutch primary vs secondary, VCUG, treat outlet first, diverticulectomy + reimplant*
- **Structural** (exstrophy, duplication, agenesis, urachal, diverticula) or **functional** (usually secondary); PBS is both.
- Every urachal anomaly = failure of allantoic obliteration (→ median umbilical ligament) at a different level.

2 · Fetal bladder on ultrasound

- Wall ≤3 mm; seen in nearly all by 13 weeks. Empties every **15–20 min** → if not seen, **repeat in the same sitting**. Amniotic fluid mainly fetal urine after **16 weeks**.
- Non-visualised bladder: absent, malformed or not filling — kidneys and amniotic fluid discriminate. Exstrophy: no filling with **normal amniotic fluid**.

	Agenesis	Hypoplasia	Exstrophy
US	Absent after repeat scan	Small, never reaches capacity	No filling; template on abdominal wall
Core defect	Cranial UGS atrophy or no trigone	Inadequate filling/storage	Open bladder plate
Outcome	Survives only with ectopic ureteric drainage	May grow; augmentation often	Reconstruction → *Ch 43*

- **Agenesis** (~1:600,000, mostly female): ureters drain to müllerian structures (female) or rectum (male); renal dysplasia/agenesis, absent genital structures → ureterosigmoidostomy or stoma.
- **Hypoplasia**: severe epispadias, UGS anomalies, bilateral renal dysplasia/agenesis, ureteric ectopia.

3 · Megacystis

- Classic congenital megacystis: **massive VUR causes the dilation** (urine recycles), not bladder-neck obstruction. Now any prenatally dilated bladder — a finding, not a diagnosis.

Gestation	Normal	Abnormal
10–14 weeks	Length ≤6 mm; <10% of CRL	**≥8 mm**; ≥13% of CRL
Later (functional)	Empties within 45 min	No emptying in 45 min
15–40 weeks	Length ≈ GA − 5 mm	> GA + 12 mm = megacystis

- **Causes**: LUTO — PUV ~50%, urethral atresia/stenosis, anterior valves; non-obstructive — PBS, MMIHS, cloacal anomalies. ~1/3 undiagnosed. M:F 8:1.
- **Aneuploidy ~1/5** (trisomy 13/18) → karyotype; outcome poor.
- Euploid 7–15 mm → most resolve; **>15 mm** → none (all obstructive). Resolution before 23 weeks → no postnatal surgery; after → most need intervention. **Persistent megacystis + oligohydramnios = LUTO.**

MMIHS

- Visceral myopathy (**ACTG2**); female ~70%; prenatally megacystis with normal/increased fluid. Exclude Hirschsprung (ganglion cells). TPN complications are the leading cause of death.
- CIC or vesicostomy → urodynamics (acontractile) → most need **lifelong CIC**. VUR uncommon.

> **KEY POINT**
> Female fetus with megacystis cannot have PUV → think MMIHS, cloacal anomaly, urethral atresia.

4 · Bladder duplication

- **Complete sagittal commonest**. External genital duplication up to 90%, lower GI up to 42%; VUR, renal ectopia/dysplasia. Bladder without urethra → ipsilateral dysplasia. Incomplete (one urethra) may need nothing.
- > **ALGORITHM**
  > relieve obstruction, prevent infection → later: both sphincters competent → join distal urethras; one incompetent → close that bladder neck, excise its urethra; vaginas joined + vulvoplasty
- **Sphincter competence, not anatomy, decides reconstruction.**

5 · Bladder diverticula

- Primary paraureteral (Hutch): observe if small; excise if VUR; dextranomer/hyaluronic acid still possible. Secondary: **treat outlet first**.
- **Syndromic** (**Ehlers–Danlos, Williams, Menkes**): multiple, posterolateral, giant; observe, resect for UTI/stones; recurrence common → vesicostomy option.

6 · Urachal anomalies

	Patent urachus	Sinus	Cyst	Vesicourachal diverticulum
Open	Whole canal	Umbilical end	Isolated segment	Bladder apex
Presents	Neonatal umbilical urine	Draining umbilicus	Infection (*S. aureus*), painful mass; commoner in adults	Incidental
Imaging	US; fistulogram/VCUG	Sinugram	US; CT/MRI if extensive	Incidental
Treat	Excise with cuff	Complete excision	Drain → excise	Only if narrow neck causes stones/UTI

- Differential: patent **omphalomesenteric duct**. Urachal adenocarcinoma unreported in children; incidental remnant — "excise all" vs "only if symptomatic" (increasingly adopted).

> **ALGORITHM**
> infected → drain + antibiotics first → infraumbilical incision → circumscribe at umbilicus (preserve) → extraperitoneal to dome → excise with **bladder cuff** → two-layer closure → catheter overnight

- Laparoscopic/robotic: feasible; intraperitoneal → risk of spill and urine leak.

7 · Acquired lesions

Lesion	Key features	Treatment
Haemangioma	Klippel–Trénaunay, Sturge–Weber; haematuria	Fulguration/Nd:YAG; partial cystectomy if large
Fibroepithelial polyp	2nd commonest paediatric bladder tumour after rhabdomyosarcoma	Endoscopic resection
Nephrogenic adenoma	Female 5:1; after infection, stones, surgery, augmentation	TUR; **80% recur** → surveillance cystoscopy
Cystitis cystica	Almost all female; mimics tumour	Long-term antibiotic prophylaxis
Eosinophilic cystitis	Male	**Biopsy mandatory**; resection + steroids, antihistamines
Haemorrhagic cystitis	Cyclophosphamide/ifosfamide; BK virus, CMV, adenovirus; radiotherapy	Mesna prevents; pentosan polysulfate

43 Exstrophy–Epispadias Complex

1 · Spectrum, epidemiology and embryology

- P3 P1 Bladder and ureter development, exstrophy–epispadias complex and its management; Describe embryology of bladder… theories for EEC… skeletal abnormalities — *spectrum, CBE 1 in 46,000, M:F 2.3:1, offspring 1 in 70; 22q11.2, ISL1, p63; IVF/progesterone/smoking/folate; failed mesodermal reinforcement of cloacal membrane, Marshall–Muecke and other theories; pelvis 12°/18° rotation, anterior pubis 30% short, diastasis 4.8 cm; MSRE stages, osteotomy indications, CPRE outline*
- Five entities: cloacal exstrophy; male and female CBE; male and female epispadias. Family recurrence ~1 in 100 (offspring risk ~500-fold).
- Rupture timing explains three phenotypes: early/extensive → bladder **and** hindgut open (cloacal); after urorectal septation → bladder only; genital tubercle only → epispadias.

2 · Anatomy of classic bladder exstrophy

- **Pelvis**: acetabular retroversion; posterior segment normal; consequences (short pendular penis, waddling gait, out-toeing) improve with growth. Significant spinal dysraphism uncommon (cf. cloacal).
- **Pelvic floor**: levator ani posterior, flattened; continent patients after closure have shortest intersymphyseal distance and deepest bladder neck → rationale for **deep placement, osteotomy, pelvic fixation**.
- **Intrasymphyseal band** (divergent urogenital diaphragm) tethers the posterior vesicourethral unit to the rami; contains **no striated sphincter**; radical incision to the levator hiatus is essential.
- Low umbilicus; omphalocele rare (common in cloacal). **Indirect inguinal hernia** common (bilateral, males) → repair at closure, explore other side; osteotomy lowers incidence.
- **Anorectum**: anus anterior. **Rectal prolapse before closure** resolves with closure; **after closure** → outlet obstruction → urgent cystoscopy.
- **Male**: **anterior corpora ~50% shorter**, ~30% wider → lengthening only apparent. Prostate never circumferential (PSA below norms). Testes normal, mostly retractile. Cavernous nerves lateral (potency retained); retrograde ejaculation may follow closure/BNR.
- **Female**: vagina short (~6 cm), orifice stenotic/anterior, **cervix in anterior vaginal wall**, bifid clitoris. Levator hiatus twice normal → **uterine prolapse** even nulliparous; predicted by adult diastasis, **not prevented by osteotomy**.
- **Bladder**: **immature, not neuropathic** (normal muscarinic receptors, type III collagen normalises after closure). Exposed plate → polyps, squamous metaplasia, cystitis glandularis → **adenocarcinoma risk**; delayed closure ↓ uroplakin, ↑ metaplasia. Compliance/stability fall after BNR → lifelong follow-up.
- Upper tracts usually normal (duplex commonest). Lateral ureteric entry without obliquity → **VUR in 100%** of closed bladders → reimplant at BNR.

3 · Variants

Variant	Feature	Management
Skin-covered (split symphysis) — commonest	Musculoskeletal defect, urinary tract intact	Formal closure at birth

Variant	Feature	Management
Duplicate exstrophy	Mucosal patch + closed pelvic bladder	Excise patch, close wall
Superior vesical fissure	Rupture only at top; sphincter intact	Toilet-trains normally; no BNR
Pseudoexstrophy	Musculoskeletal defect only; low umbilicus	—
Epispadias with bladder prolapse	May show only on Valsalva	Repair ± osteotomy

KEY POINT
Superior vesical fissure inflated the belief that variants do better. All others need formal closure and follow-up like CBE.

4 · Prenatal diagnosis

- Only ~25% prenatal; often mislabelled omphalocele/gastroschisis. Signs: **absent bladder filling**, low umbilicus, wide pubic rami, small genitalia, **enlarging lower abdominal mass**; cord insertion–genital tubercle distance <5th centile; diastasis >10 mm. Fetal MRI if CBE vs cloacal unclear → **delivery at an exstrophy centre**.

5 · Newborn care and selection

ALGORITHM
tie cord with 2-0 silk (no clamp) → non-adherent plastic wrap → saline irrigation at each nappy change → cardiopulmonary assessment, renal US → transfer to exstrophy centre; MDT counselling

- **Assess** template (≥ **3 × 3 cm** suitable), diastasis and malleability under anaesthesia, urethral plate, penile length. Plate that **bulges on crying or indents under anaesthesia** is adequate.
- **Inadequate**: <3 cm, polyp-covered, fibrotic. Precluding primary closure: penoscrotal duplication, hypoplastic bladder, bilateral hydronephrosis; ectopic bowel in plate (relative).
- Small plate → **delayed closure at 6–9 months with osteotomy** (bad closure ruins continence). No growth by 6–12 months → excise; colon conduit or ureterosigmoidostomy.
- Delayed closure: continence later and **catheterisable channel needed far more often**; capacity loss steepest beyond 9 months. Early closure → more differentiated urothelium (uroplakin II); once surface trauma stops, a small bladder with modest outlet resistance enlarges.

6 · Surgical philosophies

Repair	Stages / feature	Weakness
MSRE	Closure → epispadias 6–12 months → BNR + reimplant 4–7 years; balanced outlet grows bladder	Multiple operations
CPRE (Mitchell)	Newborn closure + penile disassembly + neck tailoring	Hypospadias; glans loss; early reimplant
Kelly	Closure → radical soft-tissue mobilisation 3–6 months (pudendal bundles freed, muscles wrapped round urethra) → hypospadias repair ~3 years; avoids osteotomy	Fistula, stenosis

Repair	Stages / feature	Weakness
Erlangen	Total repair ~8 weeks; no osteotomy (obturator-foramen coaptation); epidural 5 days	—
Mainz	Closure → sigma pouch ~2 years; bladder remnant = seminal receptacle; oral alkalinisation	Colonic malignancy

- **Common principles**: radical posterior mobilisation; tension-free closure (osteotomy if needed); strict selection for newborn closure.

KEY POINT
Erlangen, Kelly and CPRE claim to avoid BNR, yet almost all still need an outlet procedure — compressed reconstruction adds ischaemic and outlet risk without removing that need.

7 · Primary closure (MSRE stage 1)

ALGORITHM
GA + **epidural** (T10–T12) → 2-cm mucosal strip trigone → mid-shaft/vaginal orifice → incise bladder–skin junction → free from rectus, peritoneum off dome → **divide urogenital diaphragm fibres sharply to levator hiatus** → free corpora from rami → close bladder, neck, posterior urethra; **taper over 12-Fr sound** → suprapubic Malecot (4 weeks) + ureteric stents; **urethra unstented** → **No. 2 nylon horizontal mattress** pubic suture, knot away from neourethra → neoumbilicus

- **Tapering**: enough resistance for growth and to prevent prolapse, not to harm kidneys.
- **Ureteric stents** 10–14 days (oedema obstructs ureters); urethral stent → necrosis.
- **Commonest error**: inadequate urogenital diaphragm dissection → unit pulled anteriorly → prolapse.

8 · Osteotomy and immobilisation

- Aims: close ring, relieve tension, place unit deep, reduce hernias.
- **Newborn <72 h**, malleable pubes meeting on trochanter rotation → no osteotomy. **If in doubt, osteotomise.**
- Osteotomy if: closure **>72 h**, diastasis **>4 cm**, stiff pelvis, large template, **failed closure**, **cloacal exstrophy**. Diastasis >6 cm → staged closure with fixator.

Type	Position	Features
Combined bilateral anterior innominate + vertical iliac (preferred)	Supine	Best malleability; least recurrence
Bilateral posterior iliac (original)	Prone	Worse approximation, more recurrence
Anterior innominate alone	Supine	Less mobility

- Technique: Gigli-saw transverse innominate cut between anterior superior and inferior spines + vertical closing-wedge iliac cut lateral to sacroiliac joint (posterior cortex hinge) → pins → repair → pubic suture → **external fixator**.
- Aftercare: radiograph 7–10 days (crank fixator); **modified Buck traction ~4 weeks + fixator 4–6 weeks**. No osteotomy: **modified Bryant traction** (hips 90°) 4 weeks. Every pubic repair is weaker than intact symphysis → rigid immobilisation.

Immobilisation (CBE)	Failure overall	With osteotomy	Without osteotomy
Spica cast	47%	30%	**66%**
External fixation + traction	**7.5%**	8%	— (few)

- Immobilisation matters independently: failure ↓ with Buck + fixator **>4 weeks**, ↑ if non-paediatric orthopaedic surgeon; osteotomy type and epidural irrelevant.
- Complications: pin-site inflammation (near-universal), transient femoral nerve palsy (recovers by 12 weeks), delayed union. Partial diastasis recurrence (pin loosening, ischiopubic undergrowth) — still net beneficial.

9 · After primary closure

- Keys: **bladder drainage, ureteric drainage, immobilisation, pain control**. At 4 weeks: calibrate outlet, US, clamp for residual. **Continuous antibiotic prophylaxis** (all have VUR).
- US 3 months then 6–12-monthly; **yearly cystoscopy + cystogram under anaesthesia** for capacity.
- Capacity **<30 mL at 1–2 years** → continence procedure may not be feasible.
- **Outlet too tight** → dilation → incision → early reimplant → revision. Recurrent UTI + distended bladder → **intrapubic stitch erosion**. Keep residuals <50 mL.

10 · Epispadias repair (stage 2)

- Before BNR because urethroplasty **increases capacity**; 6–12 months. **Testosterone enanthate 2 mg/kg IM at 5 and 2 weeks** before every urethroplasty.
- **Modified Cantwell–Ransley** (preferred): plate freed but **distal 1 cm left on glans**, tubularised over 8 Fr, **corpora rotated over neourethra** (urethra ventral); stent 10–12 days. Remnant suspensory ligaments/scar always excised and corpora freed from rami.
- **Chordee**: dorsomedial corporotomy + cavernocavernostomy or, in adolescents, dorsal dermal graft. **Skin**: ventral prepuce as lateral flaps or buttonholed; base Z-plasty. Abandoned: in-situ dorsal tubularisation, free grafts, transverse island flaps. Adolescent males often value appearance over continence.
- Results: downward penis, catheterisable urethra; early fistula ~1 in 6 (many close by 3 months), at base — **coronal fistula rare**. Scarce tissue → tunica vaginalis flap.
- **Mitchell–Bägli disassembly**: plate ischaemia; most **hypospadiac**. Glans/corporal loss linked to closure **without osteotomy** → if ischaemia seen, immediate osteotomy.
- Control bladder spasm (epidural + oxybutynin) — spasm → extravasation, fistula.
- **CPRE**: ~half need reimplant in year 1; girls closed after 72 h risk retention; complications as frequent as MSRE but more serious. Voided continence without outlet surgery ~1 in 5; 65–80% need outlet procedure; most end on CIC.
- **Mitchell**: osteotomy if >3 days old or wide diastasis; in girls neck, urethra and vagina mobilised as one unit; short urethra → hypospadiac meatus. **Kelly** stage 2: plate beneath corpora, pubic "flake", pudendal bundles freed in ischiorectal fossa. **Kelly**: unaided voided continence <1 in 5, half partially continent; glans loss reported.
- **Combined closure + epispadias repair**: failed/delayed closure, boys >6 months; ~two-thirds dry after YDL; ischaemic penile loss risk; experts only.
- **Female closure**: osteotomy less often (>4 cm or stiff); bladder, urethra, vagina moved deep as one unit; **monsplasty, join clitoral halves**, Y–V labioplasty.

11 · Bladder neck reconstruction (stage 3)

- **Prerequisites**: capacity **≥100 mL** (under anaesthesia), stable urodynamics, motivated child (5–7 years), voiding programme ≥6 months, psychology review. Earlier reimplant for pyelonephritis, worsening hydronephrosis or falling function.

> **ALGORITHM**
> **Modified YDL**: transverse neck incision → **Cohen cross-trigonal reimplant** → posterior mucosal strip **15 × 30 mm** → denude lateral muscle triangles (no cephalad transverse cut) → tube → **double-breast muscle flaps** → suspend to rectus fascia; suprapubic tube 3 weeks, clamp trials

- MSRE: ~**70–75%** dry by day (>3 h); CPRE + YDL lowest (~56%).
- **Predictors**: template size, **successful primary closure**, **capacity at BNR** (most important; >100 mL → dry in half the time). Closure <72 h or with osteotomy → continence likely; delayed without osteotomy ~10%. After two closures <1 in 5 void continently.
- High "continence" in large cohorts mostly from **bladder-neck closure + catheterisable stoma**.
- Day-dry within ~12 months; **not dry by 2 years = failure**; night dryness 2–3 years.

Category (after BNR)	Definition
Continent	Dry day and night, interval ≥3 h
Socially continent	Dry 3 h by day, occasional wet nights
Wet	Dry <3 h by day, wet at night

12 · Failures and complications

- **Failed closure**: dehiscence, **bladder prolapse**, outlet stricture, soft-tissue loss, vesicocutaneous fistula. Causes: incomplete pelvic-diaphragm mobilisation, inadequate immobilisation, infection, distension, tube malfunction.
- Expert series: major dehiscence/prolapse ~3%; minor urological ~11%; major orthopaedic ~6% of osteotomised; femoral palsy ~2%.
- **Evaluate**: template, perivesical scarring, diastasis, prior osteotomy, ≥4 months since failure. Vesicocutaneous fistula (lower midline tract, widened diastasis) usually signals failed closure.
- **Reclosure**: wait ≥4–6 months; **osteotomy** (repeat safe), rigid immobilisation, radical urogenital diaphragm division; may combine epispadias repair after testosterone. After CPRE tissue loss → grafts/expansion first.
- Stricture (most within 60 days): UTI, enlarging bladder, stones, long dry intervals, rectal prolapse → dilation/incision → urethroplasty → diversion. Obstruction >6 months → usually augmentation.
- Secondary-closure bladders smaller; voided continence after reclosure ~1 in 6; bulking disappointing. Fellowship-trained paediatric urologists reclose better.
- **Failed BNR** (no 3-h dry interval by 2 years): near-miss (dry >2 h) → bulking; no benefit if dry <1 h. **Redo YDL** only if patulous neck, capacity >100 mL, stable urodynamics; else **neck transection + augmentation + Mitrofanoff**; stones ~30%. No pubertal "prostatic" continence.
- **Genital**: urethrocutaneous fistula commonest; glans/corporal loss → expansion, grafts, **radial forearm phalloplasty**.

13 · Diversion and augmentation

- Indications: unclosable plate, failed closure without growth, failed BNR, upper-tract decompensation.
- **Ureterosigmoidostomy**: needs anal continence; pyelonephritis, hyperchloraemic acidosis, **colonic adenocarcinoma ~3 decades later → yearly colonoscopy**. Mainz sigma pouch lowers pressure; severe renal impairment → colon conduit.
- Too small → **non-refluxing colon conduit**. Failed BNR → **augmentation + appendiceal channel ± neck closure**; stones commonest complication. Augmentation does not harm long-term GFR.
- **Continent diversion indications**: failed closures not growing after reclosure; newborn closures not growing; inadequate capacity or wetness after BNR; upper-tract decompensation. Move early if BNR capacity unlikely.

14 · Long-term issues

- **Malignancy**: augmented exstrophy bladders → multifocal adenocarcinoma; endoscopic surveillance from 5 years after reconstruction.
- **Male sexual**: libido, erection, orgasm preserved; concerns length, appearance, chordee (adolescent dorsal graft, ligament release).
- **Male fertility** impaired (verumontanum injury, retrograde ejaculation, oligo-/azoospermia; fewer BNRs → better counts); ~1/4 paternity, half via ICSI.
- **Female**: stenotic introitus → **perineal flap vaginoplasty** (cutback worsens prolapse); monsplasty (rhomboid flaps/expanders, topical oestrogen first) for split mons; **uterine prolapse** → sacral suspension, consider prophylactic suspension at adolescent reconstruction. Routine PSA/DRE in males as other men. **Pregnancy**: **planned caesarean with urologist present**.
- Psychosocial: acting-out, sexual anxiety, suicidal ideation in older males → lifelong support, transition care.

15 · Epispadias

- P3 Development of the urethra, congenital urethral anomalies and management of epispadias — *incidence, male/Davis classification, VUR rates, >3 cm → exstrophy pathway, Cantwell–Ransley then YDL at ~100 mL, female repair, continence males 47–82% / females 70–87%*

	Male	Female
Incidence	1 in 117,000	1 in 484,000
Classification	Glanular, penile, **penopubic** (~70%, incontinent)	Davis: vesicular → subsymphyseal → retrosymphyseal (incontinent)
VUR	30–40%	30–75%
Presentation	At birth	May present only with incontinence

ALGORITHM

Female (6–12 months): vertical incision through urethral wall at 9 and 3 o'clock → excise urethral roof (dorsal wedge taper) → close over 10-Fr Foley → unite clitoral halves and labia minora, mons fat over suture line → catheter 5–7 days → BNR at 4–5 years

- Treat as exstrophy (osteotomy + closure) if diastasis >3 cm, poor intrasymphyseal bar, little sphincter, small capacity. **Male**: small bladder → urethroplasty first → BNR at ~100 mL (4–5 years); good capacity → one stage; usually no osteotomy.
- Glanular never incontinent. Disassembly in isolated epispadias: no ischaemia but often hypospadias. Old plication/suspension procedures failed.
- Female continence after BNR ~**87%** — as good as CBE girls. Epispadias outperforms exstrophy (bladder never exposed).

16 · Cloacal exstrophy

- 1 in 200,000–400,000. **OEIS**: Omphalocele, Exstrophy, Imperforate anus, Spinal defects. **Two hemibladders flanking exstrophied ileocaecal bowel**, rudimentary hindgut, separated hemiphalluses, diastasis ~twice CBE.

Feature	CBE	Cloacal
Spinal defects	Uncommon	**85–100%** (lipomeningocele commonest)
Omphalocele	Rare	Near-universal
Upper-tract anomalies	Uncommon	41–60%; solitary kidney commonest
Osteotomy	Selective	Always
Continence	Mostly voiding	Mostly augmentation + CIC

- **Neurospinal**: disrupted dorsal mesenchyme; only ~1 in 5 has sacral skin sign → MRI all; medial autonomic nerves at risk; lifelong tethering watch.
- Müllerian duplication in most girls; clubfoot; short gut (~1/4). **Intravesical phallus** (46,XY) → biopsy before calling aphallia.
- **Prenatal** (~15%): non-visualised bladder, omphalocele, myelomeningocele; **"elephant trunk" sign** = prolapsed ileum.
- **Gender**: assign by **karyotype** (historic female assignment of 46,XY abandoned).
- **Bowel**: never discard hindgut — tubularise caecal plate with end colostomy; keep spare for vaginoplasty/augmentation.

> **ALGORITHM**
> Stage 1 (newborn): close omphalocele → separate caecal plate → join hemibladders → colostomy → Stage 2: bladder closure + **combined osteotomy** + fixation 6–8 weeks → 5–7 years: YDL only if >100 mL (rare), usually augmentation + continent stoma

- Combined osteotomy ~90% success, ~sixfold odds of closure, no continence gain. Diastasis >10 cm → fixator cranked 1–2 weeks first. Staged closure ~4-fold odds vs single; success predictors: staging, osteotomy, immobilisation >4 weeks, paediatric urologist.
- Primary closure fails ~31–43% (prolapse). Continence: augmentation + CIC, median two procedures. **Neurological level** determines disability. Female cervical prolapse worse than CBE → caesarean.

44 Prune-Belly Syndrome

- P1 P3 Prune Belly Syndrome. Antenatal diagnosis, pathophysiology and postnatal management; Prune belly syndrome — *triad, 1 in 29,000–44,000, 95% male; four theories, MYOCD/FLNA; pathology by organ (dysplasia ~50%, VUR ~75%, failed pump); extra-GU rates; Woodard I–III; neonatal work-up; vesicostomy, reconstruction after 3–6 months; orchidopexy ~6 months; Randolph/Ehrlich/Monfort; creatinine <0.7 mg/dL, 30% CRF*

1 · Definition and aetiology

- **Triad** (Eagle–Barrett): deficient abdominal wall muscle; bilateral intra-abdominal testes; dilated urinary tract. Incidence ~**1 in 29,000–44,000** (≈ exstrophy); **95% male**; more in twins, African ancestry, young mothers. Name given by **Osler** MCQ.
- **Survival follows the kidneys (and lungs), not the belly**: abdominal wall is the least prognostic element. Wrinkled flaccid abdomen diagnostic on sight in the neonate.
- Heredity: 1 in 23 PBS children is a twin, but most twins discordant; chromosomal associations (Turner, trisomy 13/18, monosomy 16, Beckwith–Wiedemann) exceptional. No inheritance pattern; multiplex families: smooth muscle genes **MYOCD**, **FLNA** MCQ. Theories: early urethral obstruction, lateral plate mesoderm defect, intrinsic urinary tract defect, yolk sac defect.

> **KEY POINT**
> Obstruction theory weakness: most have **no obstruction postnatally**. The tract is **dilated, collagen-rich and largely non-obstructed** — this drives management.

2 · Genitourinary pathology

- **Kidneys**: dysplasia ~**50%**, often asymmetric; worst with outlet obstruction **without urachal decompression**. Dilation **does not correlate** with dysplasia; **infection, not obstruction, is the main threat**.
- **Ureters**: distal worse; **proximal preserved** in reconstruction. VUR ~**75%**; PUJ/VUJ obstruction uncommon. **Failed pump, not blocked pipe** → stasis → infection.
- **Bladder**: massive, **thick but smooth** (no trabeculation), urachal pseudodiverticulum; urachus patent at birth **25–30%**. Splayed trigone, lateral orifices, wide neck. **Normal compliance, large capacity, poor sensation**; "unbalanced voiding"; ~half void normally but may deteriorate → reassess.
- **Prostatic hypoplasia** → dilated posterior urethra, ejaculatory failure; distal obstruction ~20%. Vasa/seminal vesicles often atretic; retrograde ejaculation.
- **Urethral atresia** lethal unless urachus patent. **Megalourethra** commoner in PBS than any syndrome: scaphoid (spongiosum only, ventral ballooning, milder) vs fusiform (cavernosa + spongiosum, whole phallus dilates, **more severe**).
- **Testes** over iliac vessels. Infant germ cells normal → **abdominal environment** destroys later potential → basis of early orchidopexy.

3 · Extragenitourinary anomalies

~**75%** have them; nearly **half premature**.

- **Abdominal wall**: medial/inferior worst; flared costal margin, weak cough; heals well.

- **Pulmonary** (>50% survivors): hypoplasia (newborn death), pneumothorax, recurrent pneumonia. **Cardiac** ~10%: PDA, ASD/VSD, Fallot.
- **GI** ≥30%: malrotation, volvulus, atresia, splenic torsion, constipation. **Orthopaedic** up to 65%: talipes (commonest), hip dysplasia, scoliosis, **lateral knee dimples**, limb hypoplasia (iliac compression).

> **KEY POINT**
> Two mechanisms: **oligohydramnios** (lung hypoplasia, limb deformity, knee dimples) and **absent abdominal wall pressure** (weak cough, atelectasis, constipation).

4 · Presentation and spectrum

- **Prenatal**: mimics PUV and megacystis–megaureter (massive bladder, hydroureteronephrosis, irregular abdominal circumference, urachal diverticulum); dilation does **not** predict function. Dilated prostatic urethra + anhydramnios → urethral atresia.

Woodard category	Kidneys / lungs	Course	Urological management
I	Dysplasia; **pulmonary hypoplasia**; oligohydramnios, Potter, urethral atresia	Stillbirth/early death (~20% perinatal mortality overall)	Supportive; simple bladder drainage
II	Full triad; minimal/unilateral dysplasia; no hypoplasia	Risk of progressive renal insufficiency	Individualised — **controversial group**
III	Incomplete/mild; no dysplasia; stable	Majority; good	Surveillance; surgery for UTI/deterioration; orchidopexy in year 1

- Abdominal wall is **not** a sorting criterion. **RUBACE**: severity score for survivors; renal the heaviest domain MCQ.
- **Pseudo-prune**: partial flaccidity, unilateral UDT; late diagnosis → recurrent UTI → **higher late renal failure**.
- **Female PBS** (~5%): normal gonads; outlet obstruction and anorectal anomalies frequent; mortality as males.

5 · Evaluation

> **ALGORITHM**
> Team → **cardiorespiratory first** (CXR) → creatinine/BUN trend (day-1 value maternal) → renal–bladder US → prophylaxis + circumcision → selective imaging

- **VCUG** only if renal insufficiency or suspected outlet obstruction, antibiotic cover, strict asepsis; VUR up to 70%.
- **DMSA** at **4–6 weeks**; MAG3/DTPA unreliable with massive dilation; MRU for anatomy. Outlet obstruction → **percutaneous suprapubic tube**.

6 · Management philosophy

	Conservative	Early reconstruction
Premise	Non-obstructive dilation, may improve	Stasis, reflux, infection drive renal loss

	Conservative	Early reconstruction
Operate when	Proven obstruction, intractable infection	Progressive dilation, high-grade VUR, recurrent upper tract infection, declining function
Best for	Category III	Selected category II

- Delay major reconstruction to **≥3–6 months** (lungs). Single-stage comprehensive surgery (reconstruction + orchidopexy + abdominoplasty + circumcision) adds no morbidity. Correct high-grade VUR.

7 · Urinary tract surgery

- **Blocksom vesicostomy** (sepsis, renal failure, atresia): excise urachal diverticulum; stoma **larger than usual** (stenosis — low IAP). **Avoid proximal ureterostomy**.
- **Ureteroplasty**: excise distal ureter, taper upper few cm, transtrigonal reimplant. PUJ kink → ureterolysis; true obstruction → pyeloureteral anastomosis.
- **Reduction cystoplasty**: capacity recurs → only for large urachal diverticulum or wider reconstruction; **CIC** empties better.
- **Internal urethrotomy** (Otis 24–30 Fr): short-term only; abandoned in infants. Atresia: progressive dilation. Megalourethra: deglove, excise/infold.
- **Circumcision** reduces UTI.

8 · Orchidopexy

- Broad mesorchium → inguinal approach fails. **Bilateral transabdominal at ~6 months** (vessels still mobilise), often with other surgery.
- Later: **Fowler–Stephens** (vessels divided, testis on vasal collaterals); **staged** (collaterals enlarge) more successful than single-stage or microvascular autotransplant. Laparoscopy: lax wall — careful port insertion.

9 · Abdominoplasty

- Mild laxity may improve; corset helps. Cosmetic/psychosocial benefit certain; functional gains (cough, defecation, emptying) plausible but confounded. From ~6 months with other surgery; infants may need postoperative ventilation.

Technique	Incision / umbilicus	Key step / note
Randolph	Transverse, 12th rib to 12th rib via pubis	Full-thickness excision of lower (worst, per EMG) wall; lateral bulge persists
Ehrlich	Vertical; umbilicus on inferior epigastric pedicle	Vest-over-pants advancement; good long-term
Monfort	Vertical ellipse; umbilicus in situ	Central fascial plate + lateral flaps; **transperitoneal** — ideal with reconstruction/orchidopexy
Dénes	Fusiform; umbilicus buttonholed	Modified Ehrlich, double-breasted
Furness	Midline, limited	Plication; **peritoneum not opened**

ALGORITHM

Monfort: skin ellipse, umbilicus spared → incise central plate at lateral rectus borders → GU surgery → score peritoneum over lateral muscle → suture plate edges to scored line → overlap lateral flaps over drains

10 · Long-term outcome

- Creatinine **<0.7 mg/dL at 1 year** → stable unless pyelonephritis/obstruction. Up to **30%** CRF (drivers: **pyelonephritis, dysplasia, polyuria**).
- **Transplant**: equal outcomes; ensure bladder emptying first.
- **Bladder**: timed, double voiding, Credé; failure → CIC. Constipation >half; asthma; musculoskeletal correction improves QoL most.
- **Fertility**: normal puberty and erections; **azoospermia, no spontaneous paternity** → sperm retrieval + ICSI. Females can carry pregnancies.

KEY POINT

Long-term care is a contest against **stasis**: lowering residual and infection protects kidneys; dysplasia and polyuria set the ceiling surgery cannot raise.

45 Posterior Urethral Valves

1 · Definition and classification

- **PUV**: commonest cause of infant male bladder outlet obstruction. Ablation relieves obstruction but **not the embryological injury** to bladder and kidney → lifelong urology–nephrology care.
- LUTO ~2.2/10,000 births; PUV commonest (~1.4/10,000), then urethral atresia, prune-belly. Commoner in Black and minority ethnic groups.
- P1 Embryology of Posterior Urethral Valves. — *Young 1919; types I–III and origin; COPUM (one membrane torn by catheter); genetics: chr 11 duplications, CNVs (worse renal outcome), TBX5, PTK7*

Young type	Anatomy	Status
I (95%)	Leaflets from verumontanum, fuse anteriorly just proximal to external sphincter	Obstructing; the lesion treated
II	Folds verumontanum → bladder neck	Non-obstructing; probably not real
III (5–10%)	Annular membrane, central aperture (persistent urogenital membrane)	Obstructing

- CNV burden predicts higher nadir creatinine and CKD 5 — links genotype to renal prognosis, not to valve presence.

2 · Pathophysiology

- Obstruction → detrusor hypertrophy → **fibrosis replacing smooth muscle** → reduced contractility. The bladder is the **driver**: renal dysfunction, VUR and voiding dysfunction all act through it.
- VCUG triad: **dilated posterior urethra, hypertrophied bladder neck, flattened verumontanum** — regresses after ablation.
- Ureteric dilation: **pressure transmission, VUR (up to 70%), polyuria**; chronic dilation → lost peristalsis/coaptation → stasis, infection.
- Renal injury = **obstructive uropathy** (partly reversible) + **dysplasia** (fixed at birth; sets the ceiling).
- Tubular injury + reduced medullary flow → **nephrogenic polyuria**, post-obstructive diuresis.
- Renin rises before creatinine; angiotensin II → TGF-β1 → fibrosis. ACEIs lower TGF-β1/TNF-α — possible role in selected children; both fall after ablation (candidate biomarkers).
- Fetal sheep: irreversible dysplasia by term — renal rescue window closes early.

Organ / lesion	Effect	Natural history
Lung — hypoplasia	Respiratory failure	Commonest cause of neonatal death
Kidney — obstructive uropathy	Reversible insufficiency	Improves; may recur with bladder dysfunction
Kidney — dysplasia	Irreversible insufficiency	Progressive failure, hypertension
Kidney — tubular injury	Salt and water wasting	Nephrogenic DI; worsens with age

Organ / lesion	Effect	Natural history
Bladder	Poor sensation, hypercontractility, low compliance → myogenic failure	Lifelong, changes with age
Ureters	Poor coaptation, stasis	Most retain hydronephrosis — **not** residual obstruction

"Pop-off" mechanisms

- **VURD** (VUR and dysplasia): unilateral high-grade VUR into a non-functioning kidney as pressure reservoir. **Refuted**: most with normal creatinine at 2 years lose it by 8–10 years; GFR abnormal early; worse urodynamics, more CIC; contralateral kidney often scarred.
- **Urinoma** and **urinary ascites**, large **bladder diverticula**: protective vs harmful debated; management same (relieve obstruction).

> **KEY POINT**
> Every pop-off is false reassurance: normal creatinine in a growing infant hides an abnormal GFR. Follow VURD as closely as any PUV child.

3 · Diagnosis

Antenatal

- US: **thick distended bladder, bilateral hydroureteronephrosis, "keyhole sign", oligohydramnios**; echogenic/cystic cortex = dysplasia. Sensitivity ~95%/specificity ~80% for LUTO, but only ~50% accurate for cause.
- Fetal MRI adjunct. ~Half diagnosed antenatally; antenatal diagnosis has **not** improved renal outcome. Postnatal: half as neonates, half with UTI.
- Male fetus, thick bladder, bilateral dilation → postnatal US and **VCUG before discharge**.

Postnatal

- **US** first: bladder wall thick if **>3 mm distended / >5 mm empty**; perirenal fluid = urinoma.
- **VCUG** (definitive): elevated bladder neck, dilated elongated posterior urethra, **abrupt funnelling** at valve; trabeculation, diverticula; high-grade VUR in ~half. 5–8 Fr feeding tube; **withdraw catheter during voiding**. ceVUS a radiation-free alternative.
- **MAG3** at **2–3 months**; slow washout from a dilated system mimics UVJ obstruction.
- Creatinine first 48 h is maternal; follow trend and **nadir** (1 month, 6 months, 1 year); plateau may be late in year 1.

4 · Fetal intervention

- Hinges on **oligohydramnios, hydronephrosis degree/laterality, gestational age**. Favourable: oligo- (not an-)hydramnios, severe hydronephrosis, normal karyotype, no cortical cysts, favourable urine.

Fetal urine (after 20 weeks)	Favourable
Sodium / chloride	**<100 / <90 mEq/L**
Osmolality	**<200 mOsm/L**
β2-microglobulin	**<6 mg/L** (most sensitive)

- **Triple serial sampling** 48–72 h apart beats one; rising values ominous. Prediction poor on systematic review. SFU (2016) standardised LUTO severity classification.
- **Vesicoamniotic shunt**: may prevent **pulmonary hypoplasia**; **no renal benefit**. Nitinol (Somatex) dislodges less than double-pigtail (Harrison).
- **Fetal cystoscopic laser ablation**: no RCT; no better than shunting.
- Both: **40–50%** pregnancy loss. Deliver at a neonatal surgical centre. Paediatric urologist counsels: diagnosis, alternatives, outcomes — start of lifelong care.

> **KEY POINT**
> Fetal intervention can rescue the lung, not the kidney: fluid deficiency is still in progress, but dysplasia is already established.

5 · Neonatal presentation and initial management

- Potter facies, club feet, weak stream, palpable bladder. **Pulmonary hypoplasia** = commonest early death and usual cause of delayed surgery (oligohydramnios + huge kidneys/bladder compress chest → lungs lack fluid volume/pressure for alveolar growth).

> **ALGORITHM**
> Stabilise airway → **5–7 Fr feeding tube** (coudé, curved stylet, or rectal finger if it curls) → confirm on US → minimal balloon fill → creatinine, electrolytes, fluids → US, then VCUG → antibiotic prophylaxis → ablation when stable

- **Post-obstructive diuresis**: large salt/water losses. †Monitor output, weight, electrolytes; replace IV.
- **Hyponatraemia** (often with UTI): transient pseudohypoaldosteronism → sodium supplements.
- **Urinoma/ascites** (3–10%): forniceal rupture or transudation/bladder rupture. Resolves with bladder drainage; tap only for respiratory distress or tense distension.
- **Delayed presentation** common (normal prenatal US): UTI, LUTS, overflow incontinence, haematuria, renal failure, rarely ejaculatory dysfunction. >40% of series present after 6 months; late ≠ milder. US then VCUG.

6 · Surgery

Primary valve ablation (preferred)

- Restores flow and **normal bladder cycling** (superior to passive diversion).
- **7.5–9 Fr cystoscope** + holmium laser or Bugbee; 9.5 Fr resectoscope with cold (Collins) knife if urethra allows (better view, meatal injury risk). Whitaker hook, Fogarty without endoscope.
- Incise at **5 and 7 o'clock ± 12 o'clock**; thin avascular → **incise, don't resect**.
- Small urethra: vesicostomy or **PADUA** — up-size urethral tube 5 → 8–10 Fr over 3–5 days.
- Catheter ≥24 h. **Repeat VCUG within 1 month**.

Complication	Cause / prevention	Management
Urethral stricture	Over-resection, hot loop, oversized instrument	†Dilation / urethrotomy; urethroplasty if dense
Meatal injury	Resectoscope through small meatus	Smaller scope / PADUA
Residual valve	Incomplete incision	†Persistent funnel on VCUG → re-ablate

Complication	Cause / prevention	Management
†Sphincter injury	Ablating distal to valves	Incontinence — precise incision

- **Bladder neck incision**: no benefit, retrograde ejaculation; bladder-neck botulinum fails — not recommended.

Vesicostomy

- For very small/premature infant, failed catheter/ablation decompression, persistent deterioration. **Does not defunctionalise**; same final outcome as ablation; bladder usable for transplant.
- P4 Vesicostomy. — *Blocksom: 2 cm transverse incision midway, dome (not anterior wall) to skin to stop prolapse; indications, complications, closure*

Upper tract diversion

- **No renal protection**; biopsies all dysplastic; complex undiversion. Only for worsening function/dilation ± sepsis **despite full lower tract drainage**, after nephrology input.

Circumcision and nephrectomy

- UTI risk 50–60% in PUV vs ~1%; circumcision cuts it >80%. Offer to all — always before reimplantation.
- VURD **nephroureterectomy** rarely (polyuria beats anuria); **keep the ureter** for augmentation.

7 · VUR in PUV

- 50–80%, **secondary**; resolves after ablation/vesicostomy in 25–40% (more if low grade, functioning kidney). Does not predict renal outcome.
- Treat the **bladder**. Reimplant only for recurrent febrile UTI despite circumcision and optimal bladder care; into an unrehabilitated bladder → stricture, persistent VUR. Endoscopic injection safer, less successful.

> **KEY POINT**
> VUR in valves is a pressure gauge, not a lesion: fix the pressure, not the reading.

8 · Bladder dysfunction and valve bladder syndrome

- P2 Valve bladder: management — *Mitchell definition; polyuria, poor sensation, residual; age pattern; continence ~5.5 y; ladder: behavioural → oxybutynin 0.1 mg/kg bd / α-blocker → CIC → overnight drainage 7–10 h → Mitrofanoff → ureterocystoplasty*

> **ALGORITHM**
> High-pressure voiding → fibrosis, weaker contraction → rising PVR + nephrogenic polyuria → upper tracts store and refill bladder → bladder never empty → pressure back-transmitted → renal decline → more polyuria

- Continence ~5.5 vs 2.9 years; history underestimates → **uroflow + PVR** in every toilet-trained boy. New incontinence after dryness = red flag.
- A continuum, not inevitable — effective drainage alters course.
- Drug logic: α-blocker for early functional bladder-neck obstruction; antimuscarinic for thick overactive wall; both harm once hypocontractile. **Mirabegron**: improves compliance, early safety data.

- **Videourodynamics** serially, especially when picture changes.
- After Mitrofanoff ~a third still deteriorate (dysplasia, polyuria). Augmentation now rare.

9 · Prognosis for renal function

- Lifetime ESRD **20–50%**. Risk: age at diagnosis, dysplasia ± VUR, **year-1 nadir creatinine**, recurrent UTI, bladder dysfunction.

Nadir creatinine	ESRD risk
1 year <0.8 mg/dL	Minimal (best single predictor)
1 year >1.2 mg/dL	High
<0.4 / 0.4–0.69 / 0.7–0.99 / >1.0 mg/dL	**0% / 2% / 27% / 100%**

- **0.7 mg/dL is the watershed** — risk leaps. 1-year beats 1-month value; 30-day creatinine velocity also predicts.
- US: echogenicity, cysts, lost corticomedullary differentiation, smaller **renal parenchymal area** → worse.
- Higher ESRD in post-US era and <1 year presenters = **survival bias**; late presenters may be more azotaemic.
- Emerging: VCUG posterior urethral ratios, deep-learning US.

10 · CKD, ESRD and transplantation → *Ch 61*

- Obstructive uropathy is a leading cause of paediatric transplantation.
- **Videourodynamics in every candidate**; optimise with CIC/overnight drainage; native nephrectomy only for proteinuria or severe polyuria.
- More **ureteric obstruction**; graft/patient survival equal if bladder managed; late graft loss linked to bladder dysfunction. Outcome same whatever initial treatment.
- Augmentation no longer "always before" — post-transplant similar; UTI risk up but graft safe with good CIC. Vesicostomy-drained bladder is a safe bridge.

11 · Long-term follow-up, sexual function and fertility

- Lifelong illness; transition to adult urologists who know valves. ADHD/neurodevelopmental disorders commoner → early assessment.
- Adult LUTS 2–3× population. ED and infertility similar overall, worse with incontinence or renal insufficiency.
- †Lifelong: BP, creatinine, urine protein, culture when febrile, US, uroflow/PVR.

12 · Related urethral lesions

- P2 Anterior urethral valve — *25–30× rarer than PUV; diverticulum; penile bulge; vesicostomy if premature, else endoscopic ablation; open repair of large diverticulum; 80% bladder dysfunction, 78% normal renal function; 25-fold risk with azotaemia+VUR+UTI*
- **Urethral atresia**: membrane at distal prostatic urethra; survives only with decompression (urachus, shunt); vesicostomy then PADUA.
- **Urethral duplication** (Effmann I blind, II complete, III caudal duplication; "Y"-type to perineum): **ventral** channel has sphincter and verumontanum → move it to glans. Blind

accessory → observe/Bugbee; two glanular openings → septum incision + dorsal urethroplasty; "Y" → staged flaps; meatus in anal sphincter → ASTRA.

- **Urethrorrhagia**: peripubertal boys, blood spots after clear urine. Incomplete sphincter relaxation/meatal stenosis → turbulence → mucosal engorgement. Up to 92% resolve → treat voiding dysfunction (history, US, uroflow, PVR). Cystoscopy only if atypical — stricture eventually in 14–28%; instrumentation may provoke it.
- Anorectal malformation: rectourinary fistula common; US + VCUG; at PSARP catheterise first, excise fistula flush, catheter ≥1 week.

13 · Landmark trials

Trial	Design	Result	Take-home
PLUTO	RCT: vesicoamniotic shunt vs conservative in LUTO	Trend to better 28-day survival; poor long-term survival and renal function in both	Shunt may help lungs, not kidneys
Circumcision RCT	Circumcision + prophylaxis vs prophylaxis	Febrile UTI 3% vs 49% over 2 years	Circumcise boys with PUV
Oxybutynin RCT	Oxybutynin to toilet training vs observation post-ablation	Better hydronephrosis/VUR; no change in creatinine, UTI, continence	Selective, monitored use

46 Neuromuscular Dysfunction of the Lower Urinary Tract in Children

1 · Definition and causes

- **Neurogenic LUT dysfunction**: bladder/sphincter dysfunction from aberrant innervation — congenital or acquired, central or peripheral, transient or permanent.
- Commonest cause: **NTD** (≈ spina bifida, SB). Survival ≈90% to 10 years → **renal failure** a leading late cause of morbidity; **10%** of SB deaths urological.
- Pivotal advance: **CIC**, Lapides 1971.

Congenital	Acquired
Myelomeningocele; occult NTD (lipomeningocele, tethered cord); sacral agenesis; ARM	Brain: CP, tumours, infarcts. Cord: trauma, transverse myelitis. Pelvis: extensive surgery

2 · Myelodysplasia (NTD)

- P1 Neurogenic bladder in myelodysplasia — *lesion levels; detrusor × outlet patterns; EAU work-up; McGuire risk; early CIC outcomes*

Classification and prevention

- **Myelomeningocele**: open, commonest and most severe, >90% lumbar/lumbosacral/sacral. **Meningocele**: meninges only. **SB occulta**: closed, up to 15% of population, usually benign.
- Neural tube closes by **day 35**. Prenatal detection: maternal AFP + US.
- **Folic acid** cuts NTDs 50–70%: **400 µg/day** for all women of child-bearing age, ≥4 weeks before conception through first month; **4–5 mg/day** if the woman has SB.
- Recurrence **5%** after one affected child; offspring 3.7% if one parent has SB (≈15% if both).
- Almost all open NTDs have **Chiari II**; most hydrocephalus (VP shunt or ETV).
- **Bony level differs from neurological level by 1–3 segments; LUT function not predictable from bony level or legs** → urodynamics for every bladder.

> **KEY POINT**
> **MOMS**: prenatal repair (<26 weeks) reduced death or shunt need (RR 0.70), better motor/cognitive scores at 30 months; cost: prematurity (34 vs 37 weeks), uterine-scar thinning/dehiscence (⅓), more re-tethering. Rationale: "two-hit" in-utero injury.

Pathophysiology

- **DSD**: sphincter fails to relax during detrusor contraction — presumed driver of remodelling and renal risk; diagnosed by EMG, not fluoroscopy.

> **KEY POINT**
> Every pattern = detrusor (over/under) × outlet (competent/incompetent/dyssynergic). Dangerous: **overactive or poorly compliant detrusor against a closed or dyssynergic outlet** → high storage pressure → VUR, hydroureteronephrosis. An incompetent outlet "pops off" — wet but kidney-safe.

Newborn evaluation

> **ALGORITHM**
> Birth → renal–bladder US before discharge + PVR (need for CIC) → creatinine after week 1 → urodynamics once stable, **safe from ≈3 months** (within 3 months: ICCS, SBA, EAU) → cystography (ideally videourodynamics, or VCUG) → ± baseline DMSA (EAU) → US at 6, 12, 18, 24 months then yearly.

- Guidelines: ICCS 2012, BAPU 2016 (most selective, "no consensus" on early CIC), SBA 2018, EAU 2022.
- UMPIRE (CDC) baseline: US normal 56%, mild hydronephrosis 40%, moderate–severe 4%; dilating VUR **7%**; DMSA defects ≈8%.
- **Cystatin C** beats creatinine (low muscle mass hides falling GFR). Gold standard: nuclear clearance.
- High-grade VUR is a pop-off → urodynamic pressures underestimate risk.

Urodynamic risk factors

Category	Definition
Hostile / high	DLPP or end-filling pressure **≥40 cmH_2O**; NDO with DSD (DSD removed in UMPIRE revision — poor agreement)
Intermediate	NDO, reduced compliance, 25–39 cmH_2O
Abnormal but safe	<25 cmH_2O
Normal	Normal capacity, compliance >15 mL/cmH_2O, no NDO/DSD, minimal residual

- Poor compliance independently predicts renal injury.

Tethered cord

- Conus below L1–L2; growth stretches cord → gait, sensory, LUT deterioration. Secondary tethering after repair common (more after fetal repair).
- † Suspect with new wetting, rising pressure/falling compliance, new UTIs, worsening scoliosis, back/leg pain.
- Urodynamics **before untethering and 3–4 months after**; needle EMG when MRI equivocal.

3 · Proactive (early) management

Priorities: **kidneys → continence → UTI avoidance → sexual function/fertility**.

- **Early CIC + antimuscarinic** (Bauer) for hostile dynamics (DSD, poor compliance, high outlet pressure); endorsed by all paediatric societies; **no benefit if the child empties spontaneously**.
- CIC bypasses DSD; easier accepted from infancy. Hydronephrosis persists → overnight indwelling drainage.
- **No** antibiotic prophylaxis on CIC; treat bacteriuria only if symptomatic.

Outcome	Expectant	Early CIC + antimuscarinic
Hydroureteronephrosis	18–80%	0–15%
Renal scarring	27–50%	4–12%
ESRD	≈30%	0–1.6%

Outcome	Expectant	Early CIC + antimuscarinic
Augmentation	—	18–24% fewer

KEY POINT
Proactive won because it treats the cause (pressure, failure to empty) before the consequence (reflux, dilatation, scarring, renal failure).

4 · Management ladder

ALGORITHM
CIC + antimuscarinic (± β3-agonist) → intradetrusor botulinum toxin → (vesicostomy as bridge if CIC impossible) → augmentation ± catheterisable channel ± outlet procedure → *Ch 48*

Pharmacotherapy

- **Antimuscarinics** first-line (M2 most numerous, M3 mediates contraction): ↑ capacity, fewer leaks. No cognitive effect (RCT).
- FDA-licensed: **oxybutynin** (≥5 y; 0.1–0.6 mg/kg/day, max 5 mg/dose; oral, patch, **intravesical** — no first-pass), **solifenacin** (≥2 y), **fesoterodine** (≥6 y, >25 kg), **mirabegron** (≥3 y; may cause hypertension).
- **α-Sympathomimetics** (ephedrine, pseudoephedrine) for low-outlet incontinence: unlicensed, minority respond.

Botulinum toxin A

- Blocks acetylcholine (and ATP, substance P) release — motor and sensory.
- GA, rigid cystoscopy, **≈10 U/kg (max 360 U)**, ≈30 sites, **trigone spared**; usual minimum age 2 y.
- Continence 65–87%; lasts 6–10 months; repeats stay effective, no tachyphylaxis. Poor pre-existing compliance → poor response.

Surgery

- **Vesicostomy**: temporary bridge when CIC unmanageable; reverses hydronephrosis/VUR, fewer UTIs.
- **Augmentation** (ileum preferred) for refractory bladder; channel if urethral CIC hard; sling/bladder-neck reconstruction/AUS for weak sphincter → *Ch 48*. B12 yearly from 2 y after ileum; cystoscopy 10–15 y after.

VUR in the neurogenic bladder

- Secondary; treating the bladder resolves 30–55%. Grade I–III emptying fully → prophylaxis; IV–V or poor emptying → CIC; poor compliance → add antimuscarinic.
- **Avoid Credé** with reactive sphincter — raises voiding pressure, worsens VUR.
- Antireflux surgery for breakthrough febrile UTI or persistent dilation despite low pressure; with CIC succeeds ≈95% (Jeffs). At augmentation **reimplant grade ≥III** (else persists in 20–47%).

5 · Bowel management †

- † Neurogenic bowel → constipation, faecal incontinence. Ladder: diet, timed toileting → laxatives ± suppositories → transanal irrigation → **Malone ACE** (often at augmentation).

Treat constipation around reconstruction.

6 · Latex allergy †

- † High sensitisation risk (repeated mucosal exposure) → intraoperative anaphylaxis. Latex-free **from birth**; latex-free condoms.

7 · Sexual function and transition

- Function **inversely proportional to neurological level**.
- **Central precocious puberty**: ≈50% of girls, 10–30% of boys.
- **Males**: ED ≈75%; ejaculation 40–75% (some retrograde); lesion S1 or below → usually normal reproduction. **Sildenafil** effective ≈80%.
- **Females**: 70–80% conceive; caesarean and late incontinence common. Avoid combined pill (thrombosis) and IUD (pelvic infection).
- SB-specific sex education; † structured **transition**.

8 · Other causes

Sacral agenesis

- † Maternal diabetes. May walk normally — **flat buttocks, short gluteal cleft**, absent sacrum; plain radiograph.
- Level does not predict bladder; ≈⅓ present late. Areflexic bladder + weak sphincter, or NDO + DSD.

Spinal cord injury

- **Spinal shock** → CIC; after ≈3–4 months **suprasacral → NDO + DSD**; cauda equina → little recovery. Urodynamics after shock.
- **Autonomic dysreflexia** (lesion **≥T6**): filling → sympathetic surge → severe hypertension, stroke, seizures. Monitor BP, α-blockers ready during urodynamics.

Transverse myelitis

- Acute retention >95%; steroids, plasma exchange, IVIG. Areflexia ≈70% → NDO 59–90%, poor compliance ≈47%, DSD 17–80%; 57–73% need CIC.
- Clinical picture does not predict risk; **late CIC → poorer compliance** → early urodynamics and CIC.

Cerebral palsy

- Coordinated UMN: continence late; urgency; NDO 30–61%, DSD ≈12%; ⅓ normal. Upper-tract risk **<5%**; urodynamics only if conservative care fails, DSD symptoms or recurrent UTI.

ARM and pelvic surgery

- ARM: cord anomaly 18–50% (more if high) → **spinal US before 3 months**; neuropathy from **cord, not PSARP**; urodynamics before and after pull-through. Currarino: tethered cord >80%.
- Pelvic splanchnic injury → underactive bladder; hypogastric → poor compliance, open bladder neck. Sacrococcygeal teratoma (35–45% neuropathic) → urodynamics before/after; pelvic rhabdomyosarcoma survivors need surveillance.

KEY POINT

Lesion level predicts pattern: **suprapontine** (CP) → coordinated overactivity, low risk; **suprasacral cord** (SCI, myelitis, tethering) → NDO + DSD, high risk; **sacral/peripheral** → underactivity. Myelodysplasia is mixed and unpredictable.

9 · Landmark trials

Trial	Design	Result	Take-home
MOMS	RCT, prenatal vs postnatal repair	Less shunting, better motor score; more prematurity, uterine dehiscence	Fetal repair in selected centres
McGuire 1981	Myelodysplasia cohort	DLPP >40 cmH_2O → upper-tract damage	40 cmH_2O = safety line
Sildenafil in SB	Double-blind placebo RCT	≈80% improved erections	First-line for ED

47 Functional Disorders of the Bladder and Bowel in Children

1 · Normal bladder and bowel function

- P1 Describe neurophysiology of lower urinary tract and various types of voiding dysfunction in children *— innervation, PAG–PMC control, maturation, ICCS functional types, flow patterns, management principles*
- Mature LUT = **storage** (normal compliance, no overactivity, closed outlet) + **voluntary coordinated emptying**.
- **Infancy**: voiding pressures often **>100 cm H_2O** and interrupted voiding normal, gone by **~2 yr**; infants arouse before voiding. Later dyscoordination = **dysfunction, not immaturity**.
- **Rectum** distension → **RAIR**: internal sphincter relaxes to "sample" while external contracts.
- Toilet training: bowel → day bladder → night bladder. Methods equivalent; **late initiation** linked to incontinence, enuresis, UTI, withholding.

> **KEY POINT**
> Bladder and bowel are one unit: shared pelvic space ("zero-sum" volume) and converging dorsal-root afferents (cross-organ sensitisation). Loaded rectum → overactivity, poor emptying; treating constipation improves LUTS; anticholinergics for the bladder can recreate the constipation that drove the symptoms.

2 · Definitions — ICCS

BBD = LUT dysfunction + bowel dysfunction (constipation and/or faecal incontinence) in a neurologically normal child.

Group	Symptom	ICCS definition
Storage	Increased frequency	**≥8 voids/day**
	Decreased frequency	**≤3 voids/day**
	Urgency	Sudden immediate need to void
	Nocturia	Waking at night to void
Voiding	Hesitancy	Difficulty initiating when ready
	Straining	Valsalva to initiate and maintain
	Intermittency	Non-continuous stream
	Retention	Cannot void despite effort; distended bladder
Other	Holding manoeuvres; incomplete emptying; recurrent UTI	Incomplete-emptying sensation counts even with minimal PVR; recurrent UTI is itself a BBD symptom

- **Daytime urinary incontinence (DUI)**: leakage awake, **≥5 yr**. **Enuresis**: intermittent leakage asleep, ≥5 yr — **MNE** (no other LUTS) vs **NMNE** (any LUTS).
- **Faecal incontinence**: **retentive** (majority; overflow around faecal mass) vs **FNRFI** (~1 in 5; no constipation; up to half also wet).
- † **EBC** = (age in years + 1) × 30 mL (ICCS, to ~12 yr).

Daytime LUT conditions

Condition	Definition / mechanism	Key features
OAB	Urgency ± frequency, nocturia, incontinence; no UTI/pathology	Urgency incontinence, holding manoeuvres
Dysfunctional voiding	Habitual sphincter/pelvic-floor contraction **during voiding**, neurologically intact; staccato flow	Learned; high PVR, UTI, secondary VUR
Underactive bladder	Raises abdominal pressure to initiate/maintain/complete voiding	Infrequent, interrupted flow, PVR; overlaps dysfunctional voiding
Voiding postponement	† Habitual deferral with holding until urgency incontinence	† Low frequency; oppositional defiant disorder
Giggle incontinence	Extensive emptying during/after laughing; LUT otherwise normal	Distinguish from laugh-provoked OAB leak, vaginal reflux, postponement
Vaginal reflux	Vagina fills retrograde during voiding	Prepubertal girl; small leak just after voiding
Extraordinary daytime frequency	† ≥ hourly small voids, no incontinence or nocturia	† Benign, self-limiting; exclude UTI, diabetes
Hinman syndrome (non-neurogenic neurogenic bladder)	End stage of dysfunctional voiding	Trabeculation, VUR, hydronephrosis, renal damage, hypertension

KEY POINT

Trap: dysfunctional voiding (neurologically intact) ≠ DSD (neurological lesion). Use "LUT dysfunction", not "voiding dysfunction". Urgency incontinence ≠ OAB: a voiding postponer also leaks with urge, and anticholinergics help briefly then worsen emptying.

3 · Clinical presentations and associations

- **Acute retention** (peak 2–4 yr): genital inflammation, instrumentation, **faecal impaction**; exclude tumour/neurology. If comfortable → **analgesia, sitz bath and/or enema before catheter** (reflex to pelvic noxious stimulus).
- **Functional constipation**: ~14% worldwide; peak 2–3 yr. **Withholding cycle**: painful stool → withholding → harder stool → more pain. Organic: ARM, Hirschsprung, coeliac, CF, hypothyroidism, hypercalcaemia, hypokalaemia, spinal, drugs.
- **Abdominal pain**: constipation ~half of acute cases. Alarm features (weight loss, GI bleeding, fever, severe diarrhoea, vomiting) → gastroenterology.
- **Genital pain** often referred from bladder (boys grab penile tip).
- **ADHD, ASD, ID**: highest BBD rates, poorer response; treating ADHD improves enuresis (not constipation); risperidone causes incontinence. Adapt urotherapy.
- Anxiety, depression, stress associate; QoL improves even with education. Childhood BBD predicts adult OAB, incontinence, IC/BPS.

4 · Evaluation

Mainly to **exclude organic disease**; children are over-investigated.

Warning signs

Finding	Points to
Blood in stool without fissure, fever, vomiting, weight loss, severe distension	Organic GI/systemic disease
Failure to thrive, raised BP	CKD
Lumbosacral dimple, hair tuft, vascular lesion, asymmetric cleft	Occult dysraphism → spinal MRI
Weak/numb legs, abnormal gait; absent anal or cremasteric reflex	Neurogenic bladder/bowel
Abnormal anal position or small anus	ARM
Low SG, glucosuria, proteinuria	DI, DM, CKD
Girls: constant dribble despite normal voiding	Ectopic ureter

- Ask **concretely** (stools this week, Bristol pictures); ask about bowels in **every** child with LUTS. **3-day bladder diary + bowel diary**. Review prior treatment, school toilets, neurological history.
- **Bristol**: 1–2 hard (constipation) · 3–4 normal · 5–7 soft → watery.
- Exam: growth, gait, back, neurology, abdomen, anogenital; **gush on separating labia** = vaginal reflux. Routine DRE **not** needed.

Investigations (stepwise, non-invasive)

- Scores: DVSS, Vancouver. **ICCS response**: complete 100%, partial 50–99%, none <50%.
- **Urinalysis**: SG >1.005 makes DI unlikely; persistent protein ≥1+ → first-morning protein:creatinine (>**0.2 mg/mg** abnormal).
- **PVR** within 5 min of voiding, bladder 50–115% EBC. Renal US not routine; DVSS ≥12 + UTI predicts abnormal US.
- **Uroflow** (3 flows, >50% EBC): bell normal, tower OAB, staccato dysfunctional voiding, interrupted underactive, plateau obstruction.
- **Pelvic floor EMG**: active during voiding = dysfunctional voiding. **EMG lag time <2 s** → detrusor overactivity.
- **Video urodynamics**: only for treatment failure or suspected obstruction/upper-tract risk.

Bowel — Rome IV functional constipation (child ≥4 yr)

> **ALGORITHM**
> ≥2 of, ≥1/week for ≥1 month, IBS criteria not met: ≤2 defecations in toilet/week · ≥1 faecal incontinence/week · retentive posturing/withholding · painful or hard stools · large rectal faecal mass · large-diameter stools that may block the toilet

- Pain is **not** a criterion; pain persisting after resolution → think IBS.
- No routine labs or **plain radiograph** (only if impaction suspected and exam impossible, or refractory). Rectal diameter **≥30 mm** on US suggests constipation (EAU/ESPU); gastroenterology advises against routine US.
- Refractory: transit study, contrast enema, **anorectal manometry** (absent RAIR → rectal biopsy for Hirschsprung), colonic manometry.

5 · Management of daytime incontinence

ALGORITHM

History + exam + urinalysis + PVR ± uroflow → exclude neurogenic, diabetes, CKD, anatomical → optimise neurodevelopmental disorder → **treat constipation first** → night-only? treat as MNE → else **standard urotherapy** (≥4 wk) → re-evaluate → **specific urotherapy** by diagnosis → refractory: botulinum toxin, neuromodulation

Standard urotherapy

- Structured cognitive-behavioural programme: information, instruction, lifestyle (fluids, diet, regular emptying), diaries, follow-up.
- Posture: hips/knees ~90°, feet supported, diaphragmatic breathing. **Timed voiding** 6–7/day anchored to events, timer watch.
- **Reward adherence, not dryness**. Cures DUI in more than half.

Dysfunctional voiding

- Learned functional obstruction → detrusor hypertrophy, overactivity → poor compliance → decompensation → VUR, hydronephrosis (Hinman). Diagnose by **staccato flow + active pelvic floor EMG**.

Renal risk	Management
High (hydronephrosis, VUR, rising creatinine)	Video urodynamics; lower storage pressure; CIC past the sphincter
Low	Urotherapy, double voiding, treat constipation, short-course prophylaxis for recurrent UTI

- **Biofeedback = first-line specific therapy** (fewer UTIs, better flow than urotherapy). **α-blockers** (doxazosin) lower PVR. Refractory: **intrasphincteric onabotulinumtoxinA** 50–300 units.

Overactive bladder

Drugs only after urotherapy and bowel management.

Agent	Dose / point
Oxybutynin (non-selective)	IR **0.3–0.6 mg/kg/day** in 2–3 doses (max 15 mg); ER 5–20 mg/day (do not crush); patch 3.9 mg/day. Only US-approved drug for paediatric non-neurogenic OAB. Dry mouth, flushing, constipation, cognition.
Tolterodine	1–2 mg bd or ER 2–4 mg; ER RCT negative (under-dosing >35 kg)
Solifenacin (M3-selective)	2.5–10 mg daily; long half-life; positive phase III RCTs (LION, LEOPARD)
Trospium (quaternary amine)	No blood–brain barrier crossing; empty stomach
Mirabegron (β3-agonist)	25–50 mg if >35 kg; approved ≥3 yr for **neurogenic** DO only; ≈ solifenacin in OAB; safe add-on

- Check PVR and stools before and during antimuscarinics.
- **Intradetrusor onabotulinumtoxinA** (refractory): **5 units/kg, max 100 units**; lasts ~12 months; retention needing CIC rare.

Neuromodulation	Technique	Verdict
Parasacral TENS	Surface electrodes S2–S3	Beat oxybutynin; true-sham RCT negative (placebo?)
TTNS/PTNS	34-gauge needle ~5 cm above medial malleolus; great-toe flexion	Weaker than parasacral TENS
SNS	Implanted lead	Off-label; helps bladder and bowel; migration, infection, revision

Other conditions

- **Underactive bladder**: timed/double voiding, biofeedback, interferential stimulation; CIC if PVR high (poorly tolerated).
- **Postponement**: timed voiding, treat comorbidity; † avoid anticholinergics.
- **Giggle**: **methylphenidate** (relapse off drug) or **biofeedback** (more durable).
- **Vaginal reflux**: void legs abducted (sit backwards), cough at end.
- **Extraordinary frequency**: † reassure.

6 · Nocturnal enuresis

- P1 P3 Nocturnal enuresis / Enuresis — *ICCS definitions, primary vs secondary (6 months dry), triad, sleep apnoea, evaluation, alarm vs desmopressin, second line*
- Prevalence **5–10% at 7 yr**; ~15%/yr resolve; ~2% adults; loci chromosomes 6 and 13.
- **Triad**: nocturnal polyuria (relative ADH deficiency), reduced nocturnal capacity/overactivity, arousal failure.
- **NMNE**: treat bothersome daytime LUTS and constipation first. **MNE**: treat from **~6 yr**.
- Ineffective: night waking, fluid restriction alone, rewards, punishment, pants.

	Enuresis alarm	Desmopressin
Use	Nightly; parent wakes child; until **14 consecutive dry nights**	**0.2–0.4 mg** tablet or 120–240 µg melt ~1 h before bed; stop if no effect ~1 week
Response	50–70%; slow	Immediate; rule of thirds; best with nocturnal polyuria
Durability	**Durable**; overlearning lowers relapse	**Not durable**; drug holidays or special nights
Safety	Nil	Hyponatraemia: † no fluids from 1 h before dose

ALGORITHM

MNE ≥6 yr → alarm or desmopressin → failure → check use → switch or **combine** → re-evaluate: hidden daytime LUTS, constipation, sleep-disordered breathing, neurodevelopmental disorder; uroflow + PVR → **add anticholinergic** to desmopressin → refractory: imipramine, β3-agonist, specialist drugs

- Sleep-disordered breathing: adenotonsillectomy helps about half. Anticholinergic response then relapse → suspect constipation.
- **Imipramine** 25–50 mg: **cardiotoxic** (overdose, long QT) → ECG if syncope, palpitations, family sudden death.

7 · Functional constipation — treatment

- P4 Management of Defecation Disorders resulting into LUTS — *why bowel causes LUTS, Rome IV, bowel first, disimpaction/maintenance PEG, biofeedback, irrigation, ACE, toxin, neuromodulation*

ALGORITHM

Impaction? → **disimpact** (oral PEG or enemas) → education, diary, toilet sitting, **maintenance PEG** → review 2 wk → failing: adherence, dose, impaction, mental health → **≥2 months** → wean when ≥1 month symptom-free, ≥3 stools/week, Rome IV not met → relapse: reinstate / gastroenterology

Laxative	Dose
PEG 3350 — disimpaction	**1–1.5 g/kg/day × 3–6 days** (>3 yr)
PEG 3350 — maintenance	**0.2–0.8 g/kg/day**, start 0.4 (>6 months)
Lactulose — maintenance	1–2 g/kg once or twice daily (safe in infants)

- Electrolyte-free PEG as effective, fewer adverse events; caution in renal impairment. **Titrate**: no stool in 24 h → increase; watery accidents → add **stimulant** (senna, bisacodyl).
- **No dependence**; wean gradually. Normal fluid/fibre (~10 g → ~20 g/day); extra water, fibre, probiotics don't help — must not delay laxatives.
- Refractory: pelvic floor biofeedback for dyssynergic defecation; transanal irrigation (¼–⅓ stop); anal sphincter toxin; SNS. **ACE**: **colonic manometry first** — diffuse dysmotility may need resection/diversion → *Ch 48*.

8 · Recurrent UTI and BBD

- RIVUR/CUTIE recurrence: **BBD + VUR 51% > BBD alone 35% > VUR alone 20%**. Wider caecal diameter → more recurrence.
- Dysfunctional voiding, postponement, underactive bladder carry more UTI than OAB; secondary VUR often resolves with treatment.
- **EAU/ESPU**: exclude BBD in every toilet-trained child with febrile/recurrent UTI; prophylaxis during BBD treatment. **AUA**: prophylaxis for BBD + VUR. ICCS: continue until symptoms improve.

9 · Landmark trials

Trial	Design	Result	Take-home
RIVUR + CUTIE (BBD analysis)	Prophylaxis vs placebo, <6 yr after febrile UTI, ± VUR	BBD + VUR highest recurrence; BBD alone > VUR alone	Treat BBD; prophylaxis for BBD + VUR
LION / LEOPARD	Phase III placebo RCTs, solifenacin	Improved all efficacy measures; mild–moderate side effects	Best paediatric antimuscarinic evidence

48 Lower Urinary Tract Reconstruction in Children

1 · Goals and principles

- Priorities: **protect kidneys → prevent UTI → continence**.
- Indication: hostile bladder (poor compliance, high pressure, poor emptying) refractory to medical care — mostly spina bifida; also exstrophy, PUV, bilateral ectopic ureters.
- **CIC** (Lapides) is the key enabler: spontaneous voiding after reconstruction is a hope, not a requirement. Main outcome determinant: **patient/family commitment**.

> **KEY POINT**
> Every reconstructive step trades storage for emptying. Raising capacity or outlet resistance turns a leaking, self-venting system into one wholly dependent on emptying. A continent uncatheterised reservoir → hydronephrosis, sepsis, perforation — incontinence can be "protective"; CIC and motivation are prerequisites.

2 · Patient selection and preoperative evaluation

- **Storage**: chronic filling pressure **>40 cm H_2O** → hydroureteronephrosis, falling GFR, acquired VUR. Anticholinergics + CIC first; persistent poor compliance → **augmentation**.
- **Outlet**: resistance must exceed bladder pressure during all activity; one of bladder neck or external sphincter suffices. Deficient → outlet procedure.
- **Emptying**: nothing predicts voiding after augmentation → every patient practises CIC preoperatively.

Urodynamic finding	Decision
Capacity + fluoroscopic shape	Bowel segment length
DLPP, end-filling pressure, open bladder neck	Bladder neck procedure
VUR + UTI history	Reimplantation
Sphincter EMG (needle more accurate)	Dyssynergia

- Fill at **≤10% of predicted capacity per minute** (fast filling falsely lowers compliance); body-temperature fluid. UMPIRE standardises care.
- US in all; hydronephrosis → renogram **with bladder catheter** (exclude primary obstruction) + VCUG/videourodynamics.
- Electrolytes, BUN, creatinine; raised → nephrology, 24-h urine incl. **volume** — reservoir must hold **3–4 h** of output.
- **Renal insufficiency**: acidosis **always worsens** with ileum/colon (consider stomach). **Liver failure**: ammonia reabsorption dangerous.
- **Reflux**: low-grade secondary usually resolves after augmentation; correct high-grade at reconstruction.
- **Bowel**: short gut (cloacal exstrophy, resection, irradiation); diarrhoea/faecal incontinence → spare ileocaecal valve. ACE at same operation if needed.
- **Route**: sensate urethra, wheelchair, prior bladder neck repair → abdominal stoma (preferred by patients). † Assess hand function, habitus.
- **Timing**: early if kidneys threatened; otherwise when incontinence socially unacceptable. All steps in **one operation** where possible.

- **Sterile urine** mandatory. Bowel prep surgeon preference; ERAS (carbohydrate to 2 h) reduces 90-day complications.
- **VP shunt**: low infection risk with sterile urine; more shunt failure; remove drains early.

3 · Managing the ureters

- Reimplant into native bladder ± **psoas hitch** (3–5 non-absorbable sutures through muscle into psoas tendon, avoid genitofemoral/femoral nerves; 8–10 cm tunnel). TUU for short ureter.
- Into bowel: **colon** — under a taenia; **stomach** — submucosal; **ileum** — needs a **nipple valve** (hardest).

4 · Bladder augmentation (enterocystoplasty)

- P1 Instestinal segments in urology — *segments compared with electrolyte disturbance; NH_4Cl absorption mechanism; other metabolic, stone, cancer complications; renal function for continent diversion*

Principles

- Cystoscopy first. **Bivalve native bladder widely** (sagittal, bladder neck to above trigone; stellate) — narrow anastomosis makes a diverticulum.
- **Always detubularise** and **reconfigure into a sphere**: intact bowel contracts at **60–100 cm H_2O**; sphere maximises volume per area.
- Length: ileum **20–40 cm**, colon **~20 cm**; err large.
- **No permanent sutures or staples** (stone nidus). Suprapubic tube via native bladder.

Segments

- **Ileocystoplasty**: at least **15–20 cm proximal to ileocaecal valve** (B12, †bile salts); open antimesenterically; U, S, W fold or cup patch.
- **Ileocaecocystoplasty**: caecum + 15–30 cm ileum. Taenial reimplant, appendix; valve loss → diarrhoea in neurogenic bowel.
- **Sigmoid**: 15–20 cm; strong contractions → detubularise (U-fold better than Mitchell).
- **Gastrocystoplasty**: † gastric wedge on gastroepiploic pedicle; for **renal insufficiency with acidosis, short gut, irradiated bowel**.
- **Jejunum**: avoided (salt-losing acidosis).

Segment	Advantages	Disadvantages
Ileum (default)	Least pressure activity; plentiful, mobile mesentery	Hyperchloraemic acidosis; B12; mucus; hard to make antireflux
Ileocaecal	Taenial reimplant; appendix; imbricated valve	Diarrhoea in neurogenic bowel; more mucus
Sigmoid	Taenial reimplant; close to bladder	Strongest contractions; most mucus; more reaugmentation
Stomach	No acidosis; least mucus; fewer struvite stones; renal failure	Hypokalaemic hypochloraemic alkalosis; haematuria–dysuria; tumours
Ureter	No mucus, no metabolic effect	Needs massively dilated ureter of non-functioning kidney; limited volume
Autoaugmentation (detrusorectomy)	Extraperitoneal; no bowel	Seldom adds capacity; compliance may not last

- **Ureterocystoplasty**: nephrectomy, ureter kept on its blood supply, opened opposite it, reconfigured into bivalved bladder. **Autoaugmentation**: detrusor excised off intact dome mucosa, edges hitched to psoas.
- Seromuscular enterocystoplasty and tissue-engineered bladder: experimental (no benefit in multicentre trial). Technique matters more than segment — use ileum unless reason not to. Robotic outcomes mirror open.

> **KEY POINT**
> Augmentation is needed less now: early anticholinergics, CIC, botulinum toxin and neuromodulation (immediate vs expectant: 17% vs 41% augmented). Not RCT-proven, but the rationale for treating at diagnosis.

Incontinent alternatives

- **Vesicostomy**: infant bridge (worsening hydronephrosis, febrile UTI with high-grade VUR, high pressure); stenosis, prolapse.
- **Ileovesicostomy**: 10–15 cm ileum, dome to rosebud stoma; catheter-free for those who cannot catheterise.

5 · Continent catheterisable channels

- P2 Discuss principles of continence in continent urinary diversion. Application of Mitrofanoff principle in continent urinary and GI reconstruction. — *flap vs nipple valve, conduit options (appendix, Monti, tapered ileum, ureter, Casale), appendicovesicostomy steps, MACE, results, stenosis*
- Flap valve: supple narrow tube tunnelled into a **low-pressure** reservoir is **compressed** as pressure rises; appendix needs only **~2 cm**. Nipple valves (Kock, Mainz, Hanna) **effaced** (~85% vs >90–95% continent). Flap valve = **no pop-off**.

> **ALGORITHM**
> Detach appendix with caecal cuff (less stenosis; caecal flap lengthens) → preserve mesoappendix → tunnel **distal end** submucosally → caecal end to skin, no tension/twist → **skin flap** into stoma → **short, straight** channel; bladder fixed to wall → test catheterisation after each step → stent 6–12 Fr 10–14 days → surgeon does first catheterisation → CIC ≥4-hourly by day

- Stoma: umbilicus or RLQ; † mark sitting, reachable by child. Appendix can be **split** for ACE + channel.
- **Yang–Monti**: 1–2 cm ileal ring retubularised **transversely** over 12 Fr; obese → double/spiral; ~2× appendix revisions. **Indiana** (imbricated ileocaecal valve): simple; diarrhoea. **Casale** bladder flap: spares bowel; uses capacity, high stenosis. Ureter/fallopian tube: more stenosis.
- **Stomal stenosis** commonest (umbilical appendix > ileum), early: L-stent + steroid cream or revision. Difficult catheterisation (redundancy, kinking) → subfascial revision. Stomal incontinence (rare): short tunnel or high reservoir pressure → urodynamics. Also polyps/granulation, caecal-extension necrosis, reservoir stones.
- Pure continent diversion (Kock ~40 cm vs ~20 cm ileum for augmentation) → keep native bladder. Orthotopic neobladder rare.
- **Ureterosigmoidostomy/Mainz II**: acidosis, pyelonephritis, cancer → largely abandoned; needs competent anal sphincter (oatmeal enema test). **Bladder buttons**: drainage only.

6 · Bladder outlet procedures

Outlet procedures sacrifice voiding and pop-off and may create a hostile bladder → neurogenic repairs need **concurrent augmentation** and surveillance **≥5 yr** (many need augmentation/toxin within 2–5 yr).

Procedure	Mechanism	Voiding	Main problem / place
Fascial sling	Coaptation + elevation	Rarely adequate → CIC	Better in girls; only predictor of success = augmentation; AUS still possible later
AUS	Inflatable bladder-neck cuff	**Preserved**	Infection, erosion, malfunction; best if voiding
Bulking injection	Submucosal	Preserved	Poor durability
Young–Dees–Leadbetter	Tubularised trigone	Possible	Exstrophy → *Ch 43*; poor in neurogenic
Kropp	Anterior bladder tube under trigone (flap valve)	Abolished	Highest continence; hard urethral CIC (boys); new VUR
Pippi Salle	Anterior flap onlaid on trigonal strip	Abolished	Easier CIC; lower night continence; fistula
Bladder neck closure	Complete separation + continent stoma	Abolished	After failed repairs; no pop-off

- **Sling** (rectus fascia 1 cm ± circumferential wrap): incise endopelvic fascia, find bladder neck by balloon traction, plane to vagina/rectum, fix to rectus sheath. Best for marginal bladder neck, deficient sphincter, child being augmented on CIC.
- **AUS**: bladder-neck cuff in girls and prepubertal boys; not outgrown. **Deactivated ~6 weeks** (pseudocapsule). Most neurogenic children still need CIC. Infection → remove. New hydronephrosis = unmasked hostility → anticholinergic/augment. Simultaneous vs staged augmentation: no difference.
- **Kropp**: 6 × 2 cm anterior flap tubularised, through trigonal tunnel. **Salle**: 1 × 4–5 cm flap, two-layer closure, wide base. Both: high delayed augmentation if not augmented.
- **Closure**: wide mobilisation, posterior lip rolled anteriorly, multilayer, **interpose omentum or rectus** → low fistula rate; only in reliable catheterisers.

> **KEY POINT**
> Choosing the outlet: voids → AUS (only option preserving voiding). On CIC and being augmented → sling (girls) or flap-valve lengthening + channel. Failed repairs → bladder neck closure + Mitrofanoff. Always re-study the bladder after.

7 · Malone antegrade continence enema (MACE)

- † Appendicocaecostomy with antireflux tunnel → scheduled antegrade washout → faecal continence (spina bifida, ARM, refractory constipation). Alternatives: Monti, caecostomy button, left-sided ACE.
- Complications: leakage, stenosis, granulation. Non-neurogenic: manometry guides selection → *Ch 47*. Plan with urinary reconstruction.

8 · Postoperative care

- Irrigate SP tube **≥3 times daily**. **Cystogram ~3 weeks** → clamp → CIC 2–3-hourly by day + 1–2 at night, **never beyond 4–5 h** by day.

9 · Complications of augmentation

Further surgery in up to ⅓ (10-year ~half) — **bowel is not bladder**.

Segment	Derangement	Management
Ileum, colon	**Hyperchloraemic metabolic acidosis** (overt mainly with renal impairment)	Bicarbonate; **base deficit** most sensitive; chlorpromazine/nicotinic acid
Jejunum	Hyponatraemic, hypochloraemic, hyperkalaemic acidosis	Avoid
Stomach	**Hypokalaemic hypochloraemic alkalosis** with vomiting; **haematuria–dysuria**	Fluids, H_2 blockers/PPI
Stomach + bowel	Near neutral	Complex patients

- Acid load → bone demineralisation, falling height percentile → treat base deficit early. B12 long term. Obstruction uncommon; diarrhoea after valve loss. † Drug reabsorption (methotrexate, phenytoin).
- **Mucus**: colon > ileum > stomach → **daily irrigation**.
- **Bacteriuria** expected with CIC; treat symptomatic infection (pain, new incontinence, fever) and **urea-splitters**; may treat haematuria, foul urine, more mucus.
- **Stones**: mostly **struvite** (bacteriuria, mucus, poor emptying — worst with abdominal stoma). Prevent: **daily irrigation ≥240 mL** saline. Stomach: uric acid.

Delayed spontaneous perforation

- Counsel **~9%**; mostly myelodysplasia; **outlet procedures double risk**; missed CIC, substance abuse. Overdistension → wall ischaemia; recurrence common.
- Also traumatic catheterisation, adhesions. Pain, distension, fever, sepsis, reduced output, shoulder-tip pain; poor sensation → **late presentation**, deaths. Gastric augments present early.
- **Cystogram** full filling + post-drain film; CT cystogram if negative. **Immediate laparotomy and repair**; conservative only if stable, sterile, contained.

Malignancy

- P1 Cancer in augmented bladder — *adenocarcinoma at anastomosis, earliest 4 yr, N-nitroso pathogenesis, gastrocystoplasty/exstrophy/transplant risk, SB 2–4%, nephrogenic adenoma, surveillance debate, cystectomy*
- Ureterosigmoidostomy adenocarcinoma at anastomosis, latency **~26 yr** (urine + faeces → N-nitroso compounds); augmentation from **4 yr**, highest after **gastrocystoplasty**, aggressive, often metastatic. SB bladder cancer ~2–4%, mostly stage III–IV. Nephrogenic adenoma (benign) commonest tumour. Annual cystoscopy: no consensus; consider in gastrocystoplasty, exstrophy, transplant. † Investigate any haematuria.

Pregnancy

- Pedicle deflects laterally. Hydronephrosis (nephrostomy over stent), UTI. Vaginal delivery possible; caesarean after major bladder neck repair or AUS; urologist present.

10 · Follow-up and transition

When	Tests
3, 6, 12 months	US; electrolytes, BUN, creatinine, culture (± urodynamics by 6 months)
Yearly, lifelong	US, chemistry (± gas, B12), CIC/irrigation compliance; urodynamics if new hydronephrosis, incontinence or VUR
Symptomatic	† Haematuria → cystoscopy; pain/sepsis → exclude perforation

- Early high-pressure contractions after augmentation → repeat urodynamics; true reaugmentation rare.
- Augmentation + bladder neck procedure → scarring, CKD ≥3 in many, driven by **CIC non-compliance**; adult urology must police CIC, irrigation, upper tracts.

11 · Landmark trials

Trial	Design	Result	Take-home
Early vs expectant neurogenic bladder care	Cohort, treated at diagnosis vs expectantly	Augmentation 17% vs 41%	Treat hostile bladder early
Daily bladder irrigation	Augmented cohort, ≥240 mL saline daily	Marked fall in stones	Routine daily irrigation
Tissue-engineered bladder	Industry multicentre trial	No gain in compliance or capacity	Bowel remains standard

★ Putting it together: common paediatric pathways

A · Antenatal UTD

ALGORITHM

Prenatal: A1 before 28 weeks → US after 32 weeks. A2–3 → US 4-weekly. Male, thick bladder, bilateral → PUV?

Postnatal: first US **after 48 hours**, except suspected PUV or bilateral severe dilation — before discharge.

Normal (APD <10 mm) → discharge · mild → repeat US · moderate–severe or dilated ureter → VCUG, MAG3, consider CAP (→ *Ch 34*).

B · First febrile UTI

ALGORITHM

Diagnose: catheter/suprapubic if not toilet-trained; bag only rules out.

Treat: within 48 h (>72 h doubles new scarring); oral if well.

Image: US all; VCUG if US abnormal, atypical or recurrent (AAP 2–24 months); DMSA at 4–6 months if scarring matters.

Prevent: treat BBD/constipation; CAP if high risk; circumcision lowers infant UTI (→ *Ch 37, 41*).

C · Newborn with PUV

ALGORITHM

Stabilise breathing → 5–7 Fr feeding tube (minimal/no balloon) → creatinine, electrolytes, fluids → US then VCUG → prophylaxis → **endoscopic ablation** when stable; vesicostomy if urethra small.

Prognosis: first-year nadir creatinine (<0.8 mg/dL good). Lifelong: valve bladder, CKD, BP; transplant → Ch 61 (book 1) (→ *Ch 45*).

D · Neurogenic bladder in spina bifida

Step	What	Why
Birth	US + PVR; creatinine after week 1; latex-free	Retention, baseline
≈3 months	Urodynamics (± video)	Hostile: DLPP or end-filling ≥40 cmH_2O, poor compliance, DSD
Hostile bladder	Early CIC + antimuscarinic	Kidney protection
Failing	Botulinum toxin → vesicostomy (infant) → augmentation ± channel ± outlet	Safe storage, emptying (→ *Ch 46, 48*)
Always	Bowel programme; re-tethering watch	Safety

E · Daytime wetting and bedwetting

ALGORITHM

Urinalysis, PVR ± uroflow → exclude neurogenic/anatomical/medical → **constipation first** → urotherapy ≥4 weeks → specific (OAB antimuscarinic, dysfunctional voiding biofeedback) → refractory: toxin, neuromodulation.

MNE: alarm or desmopressin (no fluids 1 h before to morning); combine if partial (→ *Ch 47*).

F · Augmentation segment

Segment	Advantages	Disadvantages
Ileum	Default; compliant, reliable supply	Mucus, hyperchloraemic acidosis, B12/bile-salt loss (terminal ileum)
Ileocaecum / colon	Appendix for channel; taenial reimplant	Valve loss → diarrhoea in neurogenic bowel; acidosis
Stomach	Secretes acid — renal failure; little mucus	Haematuria–dysuria, alkalosis
Ureter	Urothelium — no mucus/metabolic effect	Needs massive ureter, non-functioning kidney

KEY POINT

Reconstruction trades storage for emptying: an unreliably catheterised continent reservoir → hydronephrosis, sepsis, perforation, so CIC commitment first. Perforation = pain + sepsis; urgent (→ *Ch 48*).